

CivicWorkOS: Capability-Preserving Allocation of Municipal Work Among Humans, AI Agents and Robots

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2 ABSTRACT

Cities are moving from sensing infrastructure to agentic operation, in which AI agents perform administrative work and robots perform physical work. Allocation research optimizes fit, throughput, and ergonomics; labor research examines displacement; neither prices the developmental practice through which cities form their future practitioners. We present CivicWorkOS, which makes municipal allocation a governed decision over *staffed modes* — pairings of an execution mode with a named roster at declared career stages — rather than a human-versus-machine choice. Civic Automation Debt prices five deferred liabilities, and four constraints bind the objective: a Human Capability Preservation Budget denominated in qualified-practice hours accruing only to practitioners below independent competence, a Capability Access Constraint governing who receives protected practice, a Just Transition Constraint, and an N–1/N–2 resilience reserve. We give the city-wide program, an online Lagrangian rule with bounded between-rebalance violation, a feasibility condition, and an intake requirement for when the budget is unattainable. An exact bridge-inspection instance, reproducible by direct arithmetic from the reported values, prices protected practice at 0.0454 objective units per hour, costs 4.34% of annual objective value (0.33–9.60% across the swept parameter frontier), and yields 3.46 competent-equivalents a year against attrition of 2.88. The same arithmetic produces a result no headcount plan surfaces: machine-assisted supervised practice stretches time to competence by a factor of 3.38, from 1.2 to 4.05 years, so a city sizing intake from certification hours provisions four trainee posts where fourteen are required.

1 INTRODUCTION

Urban automation has changed character. Traditional smart-city systems sensed, connected, and analyzed urban data. Agentic systems additionally plan, decide, negotiate, invoke tools, and act, while autonomous robots physically execute those decisions in public space. Choi and Yoon Choi and Yoon (2025) demonstrate AI-agent-based urban digital twins operating over live municipal data, and Shariatpour et al. Shariatpour

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and Behzadfar (2026) describe the resulting shift toward “agentic cities” in which urban infrastructures sense, learn, predict, and act.

This creates a distributional question no existing allocation mechanism answers. A bridge inspection may be performed by a licensed inspector, a computer-vision agent, an autonomous robot, or a mixed team; a citizen service request by a public employee, an AI agent, or a human supervised by one; and sanitation, maintenance, transport, logistics, emergency response, and administration decompose the same way. Every such decision has a winner and a loser, and cities make them without any mechanism that records who lost what.

The policy problem is not whether AI and robots will replace municipal workers. It is how a city should continuously decide which tasks are automated, augmented, reserved for humans, or performed by hybrid teams, and, having decided, who performs the work that remains.

We treat that second half as constitutive. Neither automation exposure Acemoglu and Restrepo (2020); Eloundou et al. (2024) nor the developmental work that turns a novice into an expert is distributed evenly across occupational class, gender, contract status, or district. A municipality that automates its junior inspection, triage, and casework is not only reducing headcount. It is closing the route by which its next senior practitioners would have been formed, closing it first for whoever was next in line, and doing so invisibly to both a productivity objective and a post-deployment equity audit, because it operates one locally defensible task at a time.

1.1 Contributions

This article contributes the following.

- A city-scale allocation formalism whose decision variable is a *staffed mode*, a pairing of an execution mode with a roster of named practitioners at declared career stages. This is what makes the distributional questions answerable: mode selection alone cannot express who receives the work.
- *Civic Automation Debt*, pricing the deferred liability created by skill-formation loss, fallback erosion, accountability dilution, vendor dependency, and labor-transition burden. We claim no novelty for the underlying concern, which Bainbridge (1983) stated for the individual operator. The contribution is its aggregation to a municipal workforce, its formal separation from the immediate-outcome terms with which it is routinely confused, and its entry into a decision as a priced quantity with a computable flip threshold.
- A *Human Capability Preservation Budget* in qualified-practice hours that accrues only to practitioners below independent competence, with a stated unit conversion from raw certification hours, an explicit feasibility condition, and a *capability intake requirement* that tells a city how many developing practitioners the budget implies when the constraint cannot otherwise be met.
- A *Capability Access Constraint* governing who receives protected practice, so a budget sized from the incumbent workforce does not reproduce its composition for a replacement cycle, together with the equality-law and data-protection analysis that a mechanism allocating work by protected characteristic requires.
- A *Policy Digital Twin* with a rule language, status lattice, and conflict order, and *Adaptive Oversight Zones* with specified transition predicates.
- The city-wide assignment program the coupled constraints require, an online Lagrangian approximation with a bounded between-rebalance violation, and a bridge-inspection allocation carried through end to end, with a sensitivity sweep, a feasibility frontier, and two parameter shocks, all by exact arithmetic on the reported values.
- An evaluation protocol specified in full, with ten hypotheses and refutation criteria. It has not been executed, and no claim in this article rests on an outcome of it.

Section 2 positions the work and states the research questions; Section 3 defines the framework’s layers; Section 4 gives the program, algorithm, and formal results; Section 5 covers governance and lawfulness; Section 6 works the bridge-inspection allocation end to end; and Section 7 states what that establishes, the unexecuted evaluation protocol, and the limitations.

2 RELATED WORK AND RESEARCH GAP

75 This section states our positioning; Appendix S6 gives the survey work by work.

76 2.1 Task Allocation and Workforce Scheduling

77 Capability-based allocation matches task requirements to agent capabilities, so that allocation follows
 78 task-agent fit rather than automation for its own sake Ranz et al. (2017); Dorri et al. (2018). Later work
 79 added ergonomics El Makrini et al. (2019); Merlo et al. (2023), human preferences Noormohammadi-Asl
 80 et al. (2025), and measured trust Ali et al. (2022), with reviews covering robot fleets Meseguer Valenzuela
 81 and Blanes Noguera (2025) and multi-agent reinforcement learning Hady et al. (2025); the closest city-scale
 82 treatment allocates sensing tasks across a mobile crowd as a graph-structured assignment Zhao et al. (2023).
 83 All of it optimizes an operational team's performance. None asks whether repeated local decisions erode
 84 the experience that sustains future expertise, or whose expertise erodes first.

85 The capability budget of Section 3.5 belongs to the long operations-research tradition of constrained
 86 training-hour allocation rather than departing from it: nurse rostering carries skill mix, supervision ratios,
 87 and qualification maintenance as hard constraints Burke et al. (2004), and personnel scheduling includes
 88 cross-training variants Van den Bergh et al. (2013). Two differences matter. Rostering assumes the work
 89 exists and asks who covers it, whereas our decision is prior — whether a human performs it at all, and
 90 so whether developmental content exists to allocate — and rostering optimizes coverage, whereas our
 91 objective prices the future value of the practice itself.

92 2.2 Automation, Skill Erosion, and Capability Formation

93 The concern at the center of this paper is not new, and its lineage belongs on the record. Automating the
 94 easy parts of a task degrades operator skill and leaves the human unable to intervene when intervention is
 95 required Bainbridge (1983); the out-of-the-loop problem Endsley and Kiris (1995), the use, misuse, disuse,
 96 and abuse failure modes Parasuraman and Riley (1997), and the levels-of-automation scales Sheridan and
 97 Verplank (1978); Parasuraman et al. (2000) are long established, and Shneiderman Shneiderman (2020)
 98 separates automation from human control as independent axes. Zuboff's Zuboff (1988) distinction between
 99 automating and informing work is the distinction between a task that displaces a practitioner and one
 100 that develops them. Skill-decay meta-analyses Arthur et al. (1998); Tatel and Ackerman (2025) supply the
 101 instruments to calibrate D^{skill} against observed practice gaps.

102 Three consequences follow. CivicWorkOS must not claim to have discovered skill erosion. That literature
 103 works at the scale of the individual operator, whereas our question — whether a city's expertise pipeline is
 104 drawn down across thousands of independent decisions — is municipal. And it warrants treating fallback
 105 capacity separately from present competence, as D^{fall} does. The debt framing has a source too: technical
 106 debt as the deferred cost of an expedient choice Cunningham (1992); Sculley et al. (2015). Civic Automation
 107 Debt applies that accounting to a public workforce, a locally optimal allocation creating a liability settled
 108 later, by someone else, invisible on the ledger that authorized it. Rashidi Rashidi (2026) argues that AI is
 109 the first technology able to automate the entry-level cognitive work through which professional judgment
 110 is formed.

111 2.3 Fair Division and Algorithmic Due Process

112 Protected practice is a scarce, unequally distributed good, placing the Capability Access Constraint inside
 113 the fair-division literature Bouveret et al. (2016); Caragiannis et al. (2019). We use a share constraint
 114 against a policy-set target rather than an envy-based criterion because envy-freeness is defined over agents'
 115 own valuations, whereas a municipal panel must defend a distribution against a statutory equality duty
 116 and a published labor-pool composition, and because a Nash-welfare objective would depend on worker
 117 preferences no city records and workers would report strategically. The cost is that our constraint offers no
 118 envy guarantee.

119 A framework that prices civic costs must also show its working, or the pricing is unfalsifiable in the
 120 way opaque public-sector scoring is Eubanks (2018). Technological due process requires notice, a record,
 121 and a route to challenge Citron (2008); Citron and Pasquale (2014), and a right to human intervention is
 122 empty unless a human can alter the outcome Almada (2019); Vaccaro et al. (2019). Of the contestability

features Alfrink et al. (2023) synthesize, Section 5.2 implements the evidentiary record, the intervention request, and challenge to the rules themselves, and adds a systemic trigger because formally fair prioritization can still produce outcomes experienced as unjust Moon and Guha (2026). Article 22 GDPR European Parliament and Council of the European Union (2016) sets the EU floor.

2.4 Algorithmic Management, Labor, and Who Bears the Cost

AI is becoming a collaborator Senjic et al. (2026) and a manager: algorithmic management already allocates work, issues instructions, and monitors performance Syed et al. (2026); Milanez et al. (2025), with gains in consistency set against autonomy, privacy, accountability, and discrimination concerns OECD (2026) that persist even where automation performs well Wischniewski et al. (2023). Municipalities differ because public services are constituted by the discretionary judgment of street-level bureaucrats Lipsky (2010), so removing discretion changes what the service is. Ethical frameworks Callari et al. (2024), the EU AI Act's oversight requirement European Parliament and Council of the European Union (2024), and the ISO 10218 series International Organization for Standardization (2025a,b) specify principles or physical constraints, not a mechanism that turns them into real-time decisions about who performs a task.

On employment, the task-based framework Autor et al. (2003) underpins the argument that displacement is overstated when complementarities are ignored Autor (2015), and occupation-level exposure estimates Frey and Osborne (2017) fall roughly fivefold when re-measured at task level Arntz et al. (2016) — the gap between displacement management and task redesign, of which only the second is actionable by an allocation mechanism. Language models widen exposure Eloundou et al. (2024), robot adoption causally lowers employment and wages for lower-skilled workers Acemoglu and Restrepo (2020), and exposure falls unevenly: on gig and platform workers outside municipal protections Wood et al. (2019); De Stefano (2016) and on the poorest residents Eubanks (2018). The ILO just-transition guidelines specify what an institutional response must contain International Labour Organization (2015), alongside OECD policy on AI OECD (2026), employer reskilling forecasts World Economic Forum (2025), and Arab-region labor-market projections International Labour Organization and United Nations Economic and Social Commission for Western Asia (2026).

We use *capability* in the sense of Sen (1999) and Nussbaum (2011), the real freedom to do and to be that a developmental task confers and an automated one withholds, and the budget takes the institutional form Ostrom (1990) identifies for depletable shared resources: monitored, bounded, locally governed appropriation rules. None of these supplies a procedure a city can run on Tuesday morning, for one task, pricing exposure before the allocation rather than compensating for it afterwards.

2.5 Agentic-City Governance and Urban Digital Twins

Public-space robot policy concentrates on safety and privacy over wider wellbeing Mintrom et al. (2025); co-creation and transparent civic platforms raise measured trust in municipal AI Phothong et al. (2026); Nechesov and Ruponen (2024), and the workforce's own digital readiness conditions reform Aldhi et al. (2025). City digital twins already assist policy decisions Temple et al. (2024), and Díaz-Sarachaga (2025) supplies criteria for whether a deployed twin is well governed, the closest antecedent to our Policy Digital Twin. We ask the converse: whether governance can be represented inside the twin as an executable constraint on its decisions, a twin able to challenge its own model's assumptions Ben-Daya et al. (2026) and carrying the social dimensions twins usually omit Qanazi et al. (2025), which ISO 23247 Shao et al. (2023) does not cover. Resilience is a related but distinct concept, used inconsistently across the infrastructure literature Mentges et al. (2023); response networks recover faster when post-failure reassignment is planned in advance Lian et al. (2025).

2.6 Research Gap and Research Questions

These literatures remain insufficiently integrated. Table 1 positions CivicWorkOS against the closest published frameworks on the dimensions that matter for a distributional claim.

Five gaps follow. First, allocation models and policy frameworks exist separately, but no city-scale mechanism makes policy an active constraint and objective of allocation. Second, no model aggregates the single-operator mechanism of skill erosion to a municipal workforce and prices the future capability lost.

Table 1. Positioning of CivicWorkOS against the closest published allocation and governance frameworks. Columns: task–agent fit; multi-objective objective; deferred cost priced; mechanism decides who receives the work; policy enters as an executable constraint; contestability procedure specified; primary scale of application. ✓ present and specified; ▲ partially present or asserted without mechanism; ✗ absent.

Framework	Fit	Multi-obj.	Debt priced	Who works	Policy	Contest.	Scale
Ranz et al. (2017)	✓	✗	✗	✗	✗	✗	Workcell
El Makrini et al. (2019)	✓	▲	✗	✗	✗	✗	Workcell
Merlo et al. (2023)	✓	✓	✗	✗	✗	✗	Workcell
Ali et al. (2022)	✓	▲	✗	✗	✗	✗	Team
Zhao et al. (2023)	✓	✓	✗	✗	✗	✗	City sensing
Nurse rostering (Burke et al. (2004))	✓	✓	▲	✓	▲	✗	Department
Díaz-Sarachaga (2025)	✗	✗	✗	✗	▲	▲	City
Alfrink et al. (2023)	✗	✗	✗	✗	▲	✓	System
CivicWorkOS	✓	✓	✓	✓	✓	✓	City

172 Third, and most consequentially, no mechanism asks who receives the work that is preserved. A
 173 preservation rule calibrated on the current workforce reproduces its composition — gender balance,
 174 credentialing routes, neighborhoods, ratio of permanent to contracted staff — under the honorable name
 175 of preserving expertise, converting a protective mechanism into an exclusionary one. Rostering could in
 176 principle answer this, but takes the existence of the human work as given.

177 Fourth, allocation needs a measure of whether enough human capability remains to recover essential
 178 services when AI, platforms, or robots fail. Fifth, automation rules are static compliance checks, whereas
 179 the acceptable degree of automation varies with sector, risk, labor conditions, and observed performance.
 180 These gaps motivate seven research questions, addressed in Sections 3 to 7.

181 **RQ1**How can urban tasks be allocated among humans, AI agents, robots, and hybrid teams on both
 182 operational and civic objectives?

183 **RQ2**Can Civic Automation Debt price long-term capability losses without double-counting immediate
 184 outcomes?

185 **RQ3**Can a capability budget sustain expertise pipelines without fixed quotas, and at what cost?

186 **RQ4**Who receives protected developmental work, and can an access constraint stop preservation
 187 reproducing the incumbent workforce?

188 **RQ5**How can governance requirements become machine-checkable constraints with an operable
 189 contestability procedure?

190 **RQ6**Can a resilience reserve preserve critical services under automation failure?

191 **RQ7**How should oversight zones respond to performance, incidents, workforce capability, and feedback?

3 THE CIVICWORKOS FRAMEWORK

192 3.1 Problem Statement, Staffed Modes, and Civic Automation Debt

193 Consider a city containing urban tasks $\mathcal{T} = \{T_1, \dots, T_n\}$ and a heterogeneous workforce of humans (H),
 194 software AI agents (A), physical robots (R), and their hybrid combinations, giving the execution-mode set

195 $\mathcal{M}$ defined in equation (1):

$$\mathcal{M} = \{H, A, R, H+A, H+R, A+R, H+A+R\}. \quad (1)$$

196 Each task belongs to one critical service $\sigma(i) \in \mathcal{S}$ and one capability domain $\kappa(i) \in \mathcal{K}$, both taken from
197 the municipal service catalog.

198 Practitioners, Career Stage, and Staffed Modes

199 Earlier presentations of this framework selected a mode and stopped, leaving the distributional apparatus
200 indexing an object the decision variable could not express, so we make the human staffing explicit. Let $\mathcal{W}_k$
201 be the practitioners qualified in domain k , each carrying an accumulated qualified-practice total $c_w(t)$ and
202 facing a competence threshold h_k , which gives the *career stage* of equation (2):

$$\text{stage}_k(w, t) = \begin{cases} \text{developing}, & c_w(t) < h_k, \\ \text{competent}, & c_w(t) \geq h_k. \end{cases} \quad (2)$$

203 Only competent practitioners may hold roles carrying statutory sign-off. $N_k(t)$ counts competent
204 practitioners in domain k and $N_k^{\text{dev}}(t)$ the developing ones.

205 A mode m specifies $n_m \geq 0$ human slots. A *roster* a assigns those slots to distinct available practitioners
206 with a work split $\omega_{a,w} \geq 0$, $\sum_{w \in a} \omega_{a,w} = 1$; modes with no human slot admit only the null roster. The
207 pair (m, a) is a *staffed mode*, and the decision variable is defined in equation (3):

$$x_{i,m,a} \in \{0, 1\}, \quad \sum_{m \in \mathcal{M}} \sum_{a \in \mathcal{A}_{i,m}} x_{i,m,a} = 1 \quad \forall i, \quad (3)$$

208 where $\mathcal{A}_{i,m}$ is the set of feasible rosters. The *developmental eligibility* of a roster, equation (4),

$$\psi_a(t) = \sum_{w \in a} \omega_{a,w} \mathbb{1}[\text{stage}_{\kappa(i)}(w, t) = \text{developing}] \in [0, 1], \quad (4)$$

209 is the share of the human work performed by practitioners still being formed, and the *group share* of
210 equation (5),

$$\pi_{a,g} = \sum_{w \in a} \omega_{a,w} \mathbb{1}[w \in g], \quad \sum_{g \in \mathcal{G}} \pi_{a,g} = 1, \quad (5)$$

211 is well defined for a hybrid team drawn from several groups and makes the protected-practice shares of
212 Section 3.6 sum to one by construction. Worker groups $\mathcal{G}$ are a coarsening of the protected attributes
213 declared by the panel of Section 5.1 subject to a minimum cell size, not their Cartesian product
214 (Appendix S7.1).

215 Civic Automation Debt

216 Conventional optimization selects the allocation with the best combination of time, cost, accuracy,
217 or safety. The same allocation generates externalities invisible at the task level: repeated automation
218 removes entry-level learning opportunities, concentrates operational knowledge in proprietary systems,
219 reduces human fallback capacity, shifts accountability, and imposes transition costs on identifiable groups
220 of workers. The research problem is how a city can allocate work among humans, AI agents, robots,
221 and hybrid teams so as to maximize immediate performance while preserving future human capability,
222 accountability, inclusion, policy compliance, contestability, and resilience, and while remaining answerable
223 for who bears the resulting costs.

224 We define *Civic Automation Debt* (CAD) as the accumulated future liability created when automation
225 removes human practice, concentrates knowledge, reduces fallback capacity, increases dependency, or
226 displaces workers faster than the city develops alternatives. For task T_i under staffed mode (m, a) it is

quantified by equation (6):

$$CAD_{i,m,a} = \alpha D_{i,m,a}^{skill} + \beta D_{i,m}^{fall} + \gamma D_{i,m}^{acct} + \delta D_{i,m}^{dep} + \epsilon D_{i,m}^{trans}, \quad (6)$$

where D^{skill} is skill-formation loss, D^{fall} loss of human fallback capability, D^{acct} accountability dilution, D^{dep} technological dependency, and D^{trans} labor-transition burden, with weights summing to one (Section 4.3). CAD here never denotes computer-aided design.

An objection we take seriously is that these components restate quantities an allocation objective already contains, in which case the model double-counts. We separate them by measurement domain. The terms of (25) are *flows on task T_i 's own execution window* — the quality of this inspection, the cost of this dispatch — whereas the CAD components are *changes in municipal stocks* integrated over the horizon in which they settle: the pipeline, the fallback capacity, the unaccountable decision steps, vendor concentration, and workers without a reskilling pathway. The objective therefore contains no separate capability, resilience, or accountability term. Equity appears twice but as two objects: Eq^{srv} is the equity of *service delivery to residents* in this window, D^{trans} the burden on *workers* over the settlement horizon, and a city can deliver uniform service while displacing one group of workers entirely. Developmental content nonetheless reaches the decision through four distinct channels, which Appendix S7.7 separates and Section 4.7 weighs.

Each D term is normalized to $[0, 1]$ and estimable from signals a city already tracks (Appendix S7.2). Skill-formation loss is the one that carries the framework, equation (7):

$$D_{i,m,a}^{skill} = 1 - \phi_m \psi_a, \quad (7)$$

where ϕ_m is the share of the task's developmental content a human experiences under mode m (Section 3.5). Both factors are needed: a task performed entirely by machine forms nobody, and neither does one performed by an already-competent practitioner alone. Pricing the product rather than $1 - \phi_m$ is the substantive correction relative to earlier presentations. An allocation that maximizes current productivity while continuously increasing CAD draws down stocks no term in its own objective replenishes.

3.2 Layer 1: Urban Task Digital Twin

Every service is decomposed into tasks carrying a requirement vector and a duration, equation (8):

$$T_i = \{cog_i, phy_i, emp_i, risk_i, auth_i, priv_i, urg_i, learn_i, crit_i; d_i\}, \quad (8)$$

covering cognitive, physical, and social demand, safety risk, legal-authority requirement, privacy sensitivity, urgency, human learning value, criticality, and expected execution time in hours, each dimension normalized to $[0, 1]$ from sources listed in Appendix S7.3. It differs from conventional capability matching by representing the task's value for developing future competence and the duration needed to make that value budgetable. The task's *developmental content*, equation (9),

$$\ell_i = learn_i \cdot d_i \quad [\text{qualified-practice hours}], \quad (9)$$

is the unit in which the capability budget is denominated. The ISO 23247 reference architecture Shao et al. (2023) covers the sensing half of (8) but not $learn_i$ or $crit_i$, which are municipal policy judgments rather than measurements.

3.3 Layer 2: Civic Capability Ledger

The Civic Capability Ledger records, for each domain k , the state of equation (10):

$$L_k(t) = \{H_k(t), A_k(t), R_k(t), F_k(t), E_k(t), \{E_{k,g}(t)\}_{g \in \mathcal{G}}, \{c_w(t)\}_{w \in \mathcal{W}_k}\}, \quad (10)$$

comprising available human, AI, and robotic capability, fallback capacity F_k , the expertise pipeline E_k and its decomposition by worker group, without which the access constraint cannot be evaluated, and the

per-practitioner totals that (2) reads. Fallback capacity is defined in equation (11):

$$F_k(t) = \sum_{w \in \mathcal{W}_k} \zeta_w(t) \mathbb{I}[\text{stage}_k(w, t) = \text{competent}], \quad (11)$$

the qualified hours deployable under surge within a service's recovery window, with $\zeta_w(t)$ practitioner w 's surge-deployable hours net of committed work. It enters the resilience reserve through (23), and because it is conditioned on competence, a domain that stops forming practitioners watches F_k decay on the same clock as its pipeline.

3.4 Layer 3: Policy Digital Twin

The Policy Digital Twin converts laws, municipal rules, professional requirements, collective agreements, and local priorities into machine-checkable constraints. A policy rule is the tuple of equation (12):

$$\rho = \langle \text{scope}, \text{guard}, \text{status}, \text{authority}, \text{validity}, \text{provenance} \rangle, \quad (12)$$

where the scope is a conjunction of catalog predicates, the guard a decidable predicate over the task vector, roster, and ledger state, the authority ranks the source from statute down to operational guidance, the validity is a time interval, and the provenance names the instrument and the panel decision that encoded it. Statuses form the totally ordered lattice of equation (13),

$$\text{allow} \prec \text{allow-with-oversight} \prec \text{restrict} \prec \text{prohibit}, \quad (13)$$

and the twin returns the join over every firing rule, equation (14):

$$\mathcal{P}(T_i, m, a, t) = \bigsqcup_{\rho \in \text{fires}(T_i, m, a, t)} \text{status}(\rho) \in \{\text{allow}, \text{allow-with-oversight}, \text{restrict}, \text{prohibit}\}. \quad (14)$$

The most restrictive applicable rule therefore wins, the correct default for a public body, and a lower-ranked rule can never relax a higher-ranked one. The two intermediate statuses are executable: *allow-with-oversight* binds a named reviewer to the staffed mode and adds the reviewer's time and retained judgment to its terms, while *restrict* narrows $\mathcal{A}_{i,m}$ to rosters meeting a minimum human share or stage requirement, which is how a supervision ratio enters the model. Appendix S7.4 gives the conflict order, the consistency check, and a worked encoding of Article 14 of the EU AI Act European Parliament and Council of the European Union (2024); Section 5.2 makes the twin's decisions contestable.

3.5 Layer 4: Human Capability Preservation Budget

The Human Capability Preservation Budget (HCPB) reserves a minimum volume of developmental human practice, denominated throughout in *qualified-practice hours per budget period* ΔT , taken as one year, because a budget whose units do not close cannot be audited. The *human developmental share* of mode m is given in equation (15):

$$\phi_m \in [0, 1], \quad \phi_H = 1, \quad \phi_A = \phi_R = \phi_{A+R} = 0, \quad (15)$$

with hybrid values set per domain from an assessment of how much of the task's judgment a human still exercises. The HCPB constraint is equation (16):

$$\Lambda_k(\Delta T) = \sum_{i \in \mathcal{T}_k(\Delta T)} \sum_{m \in \mathcal{M}} \sum_{a \in \mathcal{A}_{i,m}} x_{i,m,a} \ell_i \phi_m \psi_a \geq B_k(t), \quad (16)$$

both sides in hours delivered to practitioners below competence. The factor ψ_a is the substantive repair: without it a domain that gives every task to its incumbents satisfies a budget sized from novice formation while forming nobody. The budget is estimated in equation (17):

$$B_k(t) = \max\left(B_k^{\text{min}}, \eta_k N_k(t) r_k(t) h_k\right), \quad (17)$$

where $r_k(t)$ is the fraction of competent practitioners lost per period, h_k the qualified-practice hours that bring one developing practitioner to competence, η_k a policy-set replacement factor ($\eta_k > 1$ builds margin, $\eta_k < 1$ deliberately lets a domain shrink), and B_k^{min} an hours floor. Certification schemes count *raw supervised clock hours* h_k^{raw} under solo unassisted practice, which accrues the mean learning value $\overline{learn}_k$ per clock hour, so equation (18) converts between them:

$$h_k = \overline{learn}_k \cdot h_k^{raw}. \quad (18)$$

A practitioner on machine-assisted supervised tasks accrues less than $\overline{learn}_k$ per clock hour, which is why realized formation takes longer than certification assumes (Section 6.6). The budget is not a jobs quota: it constrains a volume of practice rather than a number of posts, so a task may go to a developing human supported by an AI agent even where AI could perform it alone. In Ostrom's terms Ostrom (1990) it is an appropriation rule over a renewable common-pool resource, bounded, monitored, and revised by the resource's own users through Section 5.1.

3.6 Layer 4b: Capability Access and Just Transition

Equation (16) decides *how much* developmental work is protected but not *who* performs it, and left alone it has a predictable answer. Because B_k is sized from the incumbent headcount and developing practitioners are recruited through the same pipelines that produced the incumbents, a composition-blind budget allocates protected practice in proportion to who already holds the posts. A domain that is today ninety percent male, drawn from two districts, or entirely permanent while much of its labor is contracted reproduces that composition for a full replacement cycle under the name of preserving capability. We therefore separate the budget's *size*, derived from attrition, from its *distribution*, which must be set deliberately. The protected practice share of group g is defined in equation (19):

$$\Pi_{k,g} = \frac{\sum_{i \in \mathcal{T}_k(\Delta T)} \sum_m \sum_a x_{i,m,a} \ell_i \phi_m \psi_a \pi_{a,g}^{dev}}{\sum_{i \in \mathcal{T}_k(\Delta T)} \sum_m \sum_a x_{i,m,a} \ell_i \phi_m \psi_a} \quad (19)$$

where $\pi_{a,g}^{dev}$ is the group share (5) restricted to the roster's developing members, so that $\sum_g \Pi_{k,g} = 1$. The *Capability Access Constraint* is equation (20):

$$\Pi_{k,g} \geq \theta_{k,g} - \varepsilon_k \quad \forall g \in \mathcal{G}, \quad (20)$$

with target composition $\theta_{k,g}$ and tolerance ε_k . The critical design choice is that $\theta_{k,g}$ is *not* the incumbent composition. It is set by the panel of Section 5.1 from the qualified local labor pool, a statutory equality duty, or an explicit equity target, and published with the weight vector; setting it to the incumbent share remains permitted, but is then a recorded decision to freeze composition rather than an unnoticed property of the arithmetic. Feasibility requires $\sum_g (\theta_{k,g} - \varepsilon_k) \leq 1$, which the twin checks on publication. Section 5.3 addresses whether such a target may lawfully be set.

The second distributional constraint concerns those who lose work. With $\Delta_g(t, t + \Delta T)$ the net reduction in group g 's allocated hours as a fraction of its baseline, the *Just Transition Constraint* is equation (21):

$$\Delta_g(t, t + \Delta T) \leq \tau_g \quad \forall g \in \mathcal{G}, \quad (21)$$

a ceiling on the pace of displacement for any single group, following the ILO guidelines International Labour Organization (2015). It stops the framework from discovering that the cheapest way to satisfy every other constraint is to concentrate displacement on the group with least bargaining power. Reskilling coverage is a municipal budget fact rather than a function of $\mathbf{x}$, so it is enforced in the governance layer as a gating predicate rather than in the program (Appendix S7.5). Together, (20) and (21) do not resolve who wins and who loses, but make the answer explicit, versioned, auditable, and contestable.

3.7 Layer 5: Reversibility and Resilience Reserve

For each critical service s the city maintains a minimum recoverable capacity that does not depend on a single automation stack. Total service capacity comprises the automation-dependent components C_s^{aut} (an

AI agent pool, a robot fleet class, a vendor platform) and the human contribution of the domains serving s , equation (22):

$$C_s(t) = \sum_{c \in \mathcal{C}_s^{aut}} C_{s,c}(t) + \sum_{k \in \mathcal{K}_s} \xi_{s,k} F_k(t), \quad (22)$$

with $\xi_{s,k}$ the share of domain k 's surge-deployable hours that s may draw. Surviving capacity after losing the n largest automation-dependent components $\mathcal{C}_s^{(n)}$ is equation (23):

$$Res_s^{(n)}(t) = C_s(t) - \sum_{c \in \mathcal{C}_s^{(n)}(t)} C_{s,c}(t), \quad (23)$$

so what survives an AI-service outage or a fleet compromise is exactly the human capacity in the second term of (22). The *Reversibility and Resilience Reserve* constraint is equation (24):

$$Res_s^{(n_s)}(t) \geq \rho_s, \quad \rho_s = \kappa_s D_s^{peak}, \quad (24)$$

with contingency order $n_s \in \{1, 2\}$, peak demand D_s^{peak} , and $\kappa_s \in (0, 1]$ the share of that peak which must remain servable. This is the N–1 and N–2 criterion as reliability engineering states it — the system must still meet *demand* after losing its largest contributors Jain et al. (2024); Ouyang (2014) — applied to a workforce of humans, AI agents, and robots. Because F_k decays when a domain stops forming practitioners, the resilience and capability constraints are coupled, and neither can be evaluated by removing the other.

3.8 Layer 6: Policy-Aware Multi-Agent Allocation Market

Ten specialized agents price candidate staffed modes: five *capability agents* estimate what a staffed mode can do, and five *civic agents* estimate what it costs the city in stocks it must later replenish, among them the Policy Guardian Agent, which executes (14) and holds veto authority (Figure 1). The two-family split is the architectural expression of the flow and stock separation above. The market is a single-round sealed-bid call rather than an iterated negotiation, so it terminates in one round with bounded queries and is a structured elicitation mechanism rather than a strategic game; we make no equilibrium claim for it. Appendix S7.8 lists the agents and the protocol, and Section 7.5 records the principal–agent exposure the design leaves open.

4 ARCHITECTURE, OPTIMIZATION PROGRAM, AND ALGORITHM

4.1 Reference Architecture

Figure 1 realizes the layers of Section 3 as a deployable, event-driven system. The Urban Task Digital Twin publishes each task to the Civic Capability Ledger, which supplies $B_k(t)$, the practice totals $c_w(t)$, and the fallback capacity $F_k(t)$, and to the Policy Digital Twin, which supplies the policy parameters and roster guards. The Allocation Market prices candidate staffed modes and dispatches work, and a single feedback bus returns outcomes, incidents, and appeals from all three execution channels to both governance services. Robot execution sits behind fleet middleware enforcing ISO 10218-2 International Organization for Standardization (2025b), with the robots conforming to ISO 10218-1 International Organization for Standardization (2025a), so a policy-level error cannot defeat a hardware-level safety interlock.

4.2 Runtime Allocation Workflow

Figure 2 shows the per-task control flow that Algorithm 1 formalizes. If policy filtering or roster pruning empties the candidate set, the task routes to Zone Z4 human-reserved execution and the event is counted, because its rate is the operational price of the constraint system.

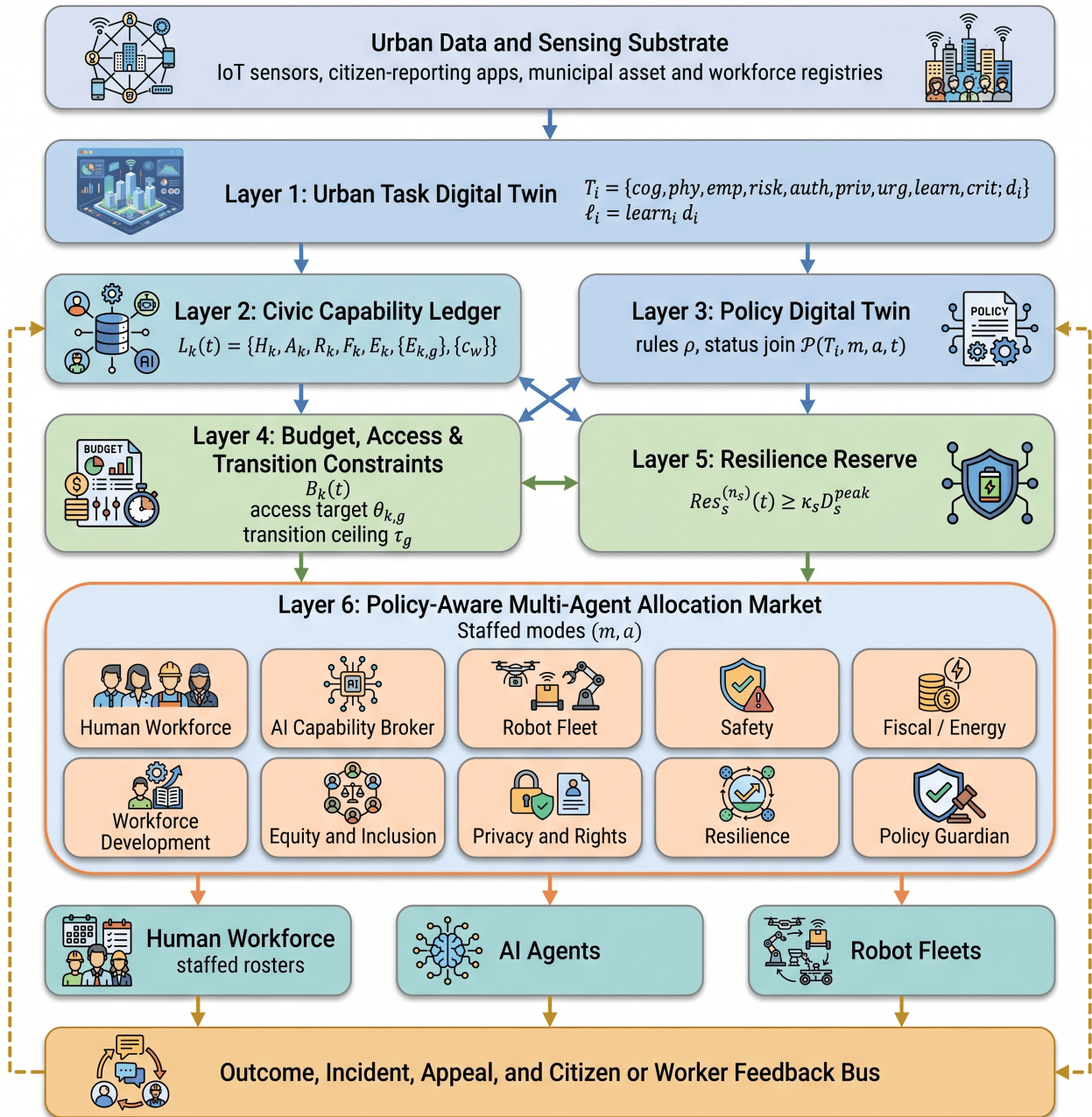


Figure 1. Reference architecture of CivicWorkOS, read top to bottom. Solid arrows are the forward allocation path from sensed urban data through the governance layers to execution; the ten agents are drawn inside the market because they constitute it. The Ledger supplies fallback capacity F_k of Equation (11) to the reserve through Equation (22), and per-practitioner practice totals c_w to the career-stage test of Equation (2). The Policy Twin supplies the replacement factor η_k , the access target $\theta_{k,g}$, and the roster guards that restrict $\mathcal{A}_{i,m}$. Dashed arrows are the outcome-feedback path, which updates both governance services.

4.3 Sustainable Civic Value (SCV)

For each task T_i and staffed mode (m, a) , the *Sustainable Civic Value* is given by equation (25):

$$SCV_{i,m,a} = w_1 Q_{i,m,a} + w_2 S_{i,m,a} + w_3 P_{i,m,a} + w_4 Eq_{i,m,a}^{srv} + w_5 Tr_{i,m,a} - w_6 Cost_{i,m,a} - w_7 En_{i,m} - w_8 Pr_{i,m} - w_9 CAD_{i,m,a}, \quad (25)$$

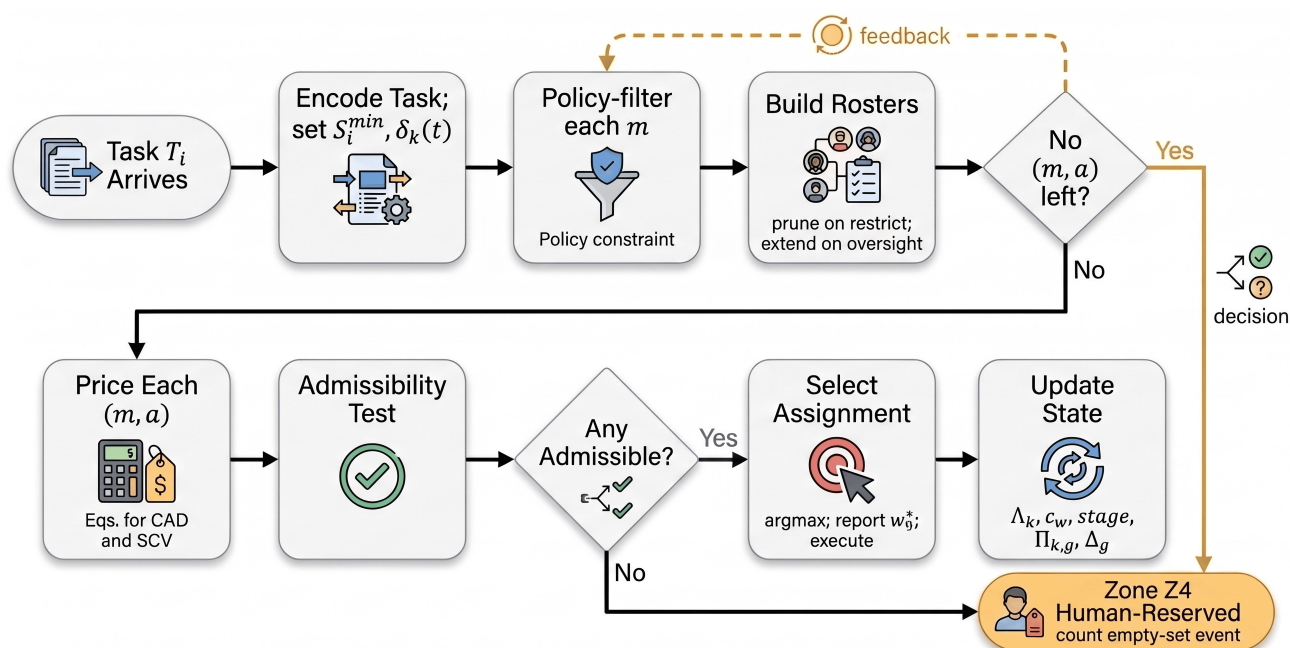


Figure 2. Runtime allocation workflow, matching Algorithm 1. The upper row encodes the task, sets the safety floor and the catch-up increment, policy-filters modes, and builds the roster set, pruning it where the twin returns *restrict* and extending it with a named reviewer where it returns *allow-with-oversight*. The lower row prices the surviving staffed modes, applies the admissibility test (29), and either falls back to Zone Z4 or selects and executes. The dashed edge returns outcomes to the Policy Digital Twin, which is what makes the zone-transition predicates (Section 4.8) evaluable.

where Q is service quality, S safety, P productivity, Eq^{srv} the equity of service delivery across resident groups, Tr agent trust, $Cost$ operating cost, En energy use, Pr privacy risk, and CAD the debt of (6). The first eight terms are flows on the task's execution window and the ninth a change in municipal stocks; six depend on the roster as well as the mode, because staffing a developing practitioner beside a competent lead changes the execution as well as its developmental content.

A linear weighting trades flexibility for interpretability relative to Pareto-front methods Osika et al. (2023); Zheng and Rasheed (2026), and recovers only solutions on the convex hull of the attainable set, a restriction Section 7.6 records.

Normalization

All nine terms are rescaled to $[0, 1]$. The eight flow terms use min-max normalization against a trailing twelve-month, sector-specific window. The debt components cannot, because against a rolling window a decaying pipeline would keep scoring mid-range and the instrument could not see the decline it exists to measure; each is instead a ratio with an externally fixed denominator (Appendix S8.2).

Default Weights and What They Actually Carry

The illustrative weights are $w_1=0.15$ (quality), $w_2=0.18$ (safety), $w_3=0.14$ (productivity), $w_4=0.10$ (service equity), $w_5=0.05$ (trust), $w_6=0.08$ (cost), $w_7=0.03$ (energy), $w_8=0.05$ (privacy risk), and $w_9=0.22$ (CAD), with debt weights $\alpha=0.28$, $\beta=0.22$, $\gamma=0.18$, $\delta=0.12$, and $\epsilon=0.20$. They are design defaults for the panel to revise, not fitted estimates; Table S3 assigns a default to every remaining parameter.

Table 2 reports the *effective* weight each construct carries once the weights are composed, because a weight vector can assert a priority the arithmetic withholds. The deferred block carries 0.22, more than productivity at 0.14, but no debt component is the largest single construct, which is safety at 0.180: the claim that CAD carries the largest weight is true of the block and false of every component. Labor-transition burden carries only 0.044, and deliberately so. An objective weight is a quantity other terms can outvote,

so transition is also carried by the hard constraint (21), which they cannot; a city that judges displacement unacceptable at any weight lowers τ_g rather than raising $w_9\epsilon$. Placing the strong lever in the constraint and the weak one in the objective is the design, and Proposition 3 shows which is deciding.

Table 2. Effective weight of each construct in the CivicWorkOS default configuration, obtained by composing $w_9 = 0.22$ with the debt weights. The deferred block is the largest single block of the objective; safety is the largest single construct. Column sums to 1.000.

Construct	Enters via	Effective weight
Safety	w_2	0.180
Service quality	w_1	0.150
Productivity	w_3	0.140
Service equity (residents)	w_4	0.100
Operating cost	w_6	0.080
Skill formation	$w_9\alpha$	0.062
Trust	w_5	0.050
Privacy risk	w_8	0.050
Fallback capacity	$w_9\beta$	0.048
Labor-transition burden (workers)	$w_9\epsilon$	0.044
Accountability	$w_9\gamma$	0.040
Energy	w_7	0.030
Vendor dependency	$w_9\delta$	0.026
Deferred block total	w_9	0.220

4.4 The City-Wide Allocation Program

The constraints of Sections 3.5 through 3.7 couple decisions across tasks — whether one inspection may go to an unstaffed autonomous team depends on how much practice the domain's other inspections have delivered, and to whom — so the program a per-task rule approximates must be written down. Over a rebalancing window, CivicWorkOS solves program (26):

$$\max_{\mathbf{x}, \varsigma} \sum_{i \in \mathcal{T}(\Delta T)} \sum_{m \in \mathcal{M}} \sum_{a \in \mathcal{A}_{i,m}} x_{i,m,a} SCV_{i,m,a} - M \sum_{k,g} \varsigma_{k,g} \quad (26a)$$

$$\text{s.t.} \quad \sum_m \sum_a x_{i,m,a} = 1, \quad \forall i, \quad (26b)$$

$$x_{i,m,a} = 0 \text{ if } \mathcal{P}(T_i, m, a, t) = \text{prohibit}, \quad (26c)$$

$$x_{i,m,a} = 0 \text{ if } S_{i,m,a} < S_i^{\min}, \quad (26d)$$

$$\sum_{i \in \mathcal{T}_k} \sum_m \sum_a x_{i,m,a} \ell_i \phi_m \psi_a \geq B_k(t), \quad \forall k, \quad (26e)$$

$$\Pi_{k,g}(\mathbf{x}) + \varsigma_{k,g} \geq \theta_{k,g} - \epsilon_k, \quad \varsigma_{k,g} \geq 0, \quad \forall k, g, \quad (26f)$$

$$\Delta_g(\mathbf{x}) \leq \tau_g, \quad \forall g, \quad (26g)$$

$$Res_s^{(n_s)}(\mathbf{x}) \geq \kappa_s D_s^{\text{peak}}, \quad \forall s, \quad (26h)$$

$$\sum_i \sum_m \sum_a x_{i,m,a} u_{i,m,a,r} \leq U_r, \quad \forall r, \quad (26i)$$

$$x_{i,m,a} \in \{0, 1\}. \quad (26j)$$

Here $u_{i,m,a,r}$ is the consumption of resource r (human hours by grade and stage, fleet hours, inference budget) and U_r its availability. The access constraint carries an elastic slack penalized at a large M , so an unattainable target yields a published shortfall rather than infeasibility and never compels an individual assignment, which Section 5.3 shows to be a legal requirement. The program is a binary multi-dimensional assignment problem with side constraints, NP-hard in general, and is therefore solved periodically over a window; this article makes no scalability claim.

4.5 Online Allocation, Duals, and the Admissibility Test

Between rebalances, tasks must be decided on arrival, so CivicWorkOS approximates (26) online by Lagrangian relaxation of all five coupling constraints, with dual prices $\lambda_k, \mu_s, \nu_{k,g}, \varrho_g, v_r \geq 0$ for (26e), (26h), (26f), (26g), and (26i) from the most recent solution. The online score is equation (27):

$$\begin{aligned} \widetilde{SCV}_{i,m,a} = & SCV_{i,m,a} + \lambda_{\kappa(i)} \ell_i \phi_m \psi_a + \mu_{\sigma(i)} \Delta_{\sigma(i)}^{res}(m, a) + \sum_g \nu_{\kappa(i),g} \ell_i \phi_m \psi_a \pi_{a,g}^{dev} \\ & - \sum_g \varrho_g \delta_g^{disp}(i, m, a) - \sum_r v_r u_{i,m,a,r}. \end{aligned} \quad (27)$$

Earlier presentations priced only the first three terms, leaving the framework blind between rebalances to the displacement it exists to bound. Because one task does not change a fleet's capacity, the resilience term is counterfactual and scaled by the task's share of service demand, equation (28):

$$\Delta_s^{res}(m, a) = \frac{d_i}{D_s^{tot}(\Delta T)} \left[Res_s^{(n_s)}(t | (m, a) \text{ generalized across } s) - Res_s^{(n_s)}(t) \right], \quad (28)$$

where $D_s^{tot}(\Delta T)$ is the service's total task-hours in the window. The duals come from the root LP relaxation, the standard construction for Lagrangian decomposition, so (27) approximates the relaxed program with error bounded by the integrality gap the solver reports (Appendix S8.3).

The budget and reserve constraints concern the state of a domain, not of a staffed mode, so testing them inside a loop over (m, a) removes every candidate or none, and a city *below* threshold would route every task to human-reserved execution, failing hardest in the state the framework exists to prevent. We therefore test the post-execution state and add a feasibility-restoration branch. Staffed mode (m, a) is *admissible* for task i at time t if it satisfies equation (29):

$$\begin{aligned} \mathcal{P}(T_i, m, a, t) \neq \textit{prohibit} \quad \text{and} \quad S_{i,m,a} \geq S_i^{min}, \quad \text{and} \\ \left[Res_s^{(n_s)}(t^+ | m, a) \geq \rho_s \quad \text{or} \quad (Res_s^{(n_s)}(t) < \rho_s \wedge \Delta_s^{res}(m, a) \geq 0) \right], \quad \text{and} \\ \left[\Lambda_k(t^+ | m, a) \geq \bar{B}_k(t) \quad \text{or} \quad \ell_i \phi_m \psi_a \geq \delta_k(t) \right], \quad \text{and} \\ \Delta_g(t^+ | m, a) \leq \tau_g \quad \forall g, \quad \text{and} \quad u_{i,m,a,r} \leq U_r - U_r^{used}(t) \quad \forall r, \end{aligned} \quad (29)$$

with $s = \sigma(i)$, $k = \kappa(i)$, where $\Lambda_k(t)$ is the practice accrued in domain k so far in the current budget period, $\bar{B}_k(t) = B_k(t) \cdot t / \Delta T$ the pro-rated budget trajectory, and the catch-up threshold $\delta_k(t)$ is defined in equation (30):

$$\delta_k(t) = \frac{\max(0, B_k(t) - \Lambda_k(t))}{n_k^{rem}(t)} \quad (30)$$

the per-task increment that closes the remaining gap over the $n_k^{rem}(t)$ tasks still expected in the period. Admitting any mode with $\phi_m \psi_a > 0$ would let a lagging domain stay behind indefinitely; requiring $\ell_i \phi_m \psi_a \geq \delta_k(t)$ makes each mode carry its share of the shortfall, so the test tightens as the gap grows. The allocation is then determined by equation (31):

$$(m, a)_i^* = \arg \max_{(m,a) \in \mathcal{A}(i,t)} \widetilde{SCV}_{i,m,a}, \quad (31)$$

over the admissible set $\mathcal{A}(i, t)$, with the Zone Z4 fallback when $\mathcal{A}(i, t) = \emptyset$. After execution, outcomes are folded back into policy through equation (32), the closed governance loop of Section 5.1:

$$Policy_{t+1} = f(Policy_t, Performance_t, Incidents_t, CAD_t, Distribution_t, Feedback_t), \quad (32)$$

Algorithm 1 states the per-task procedure, including the roster construction and the two intermediate policy statuses that earlier presentations described but never implemented.

Algorithm 1 Online policy-aware allocation over staffed modes

Require: Task T_i ; time t ; ledger $L(t)$; policy twin $\mathcal{P}$; budgets $B(t)$; thresholds ρ ; duals $\lambda, \mu, \nu, \varrho, v$ from the last solution of (26)

Ensure: Selected staffed mode $(m, a)_i^*$

- 1: Encode T_i via (8); $\ell_i \leftarrow \text{learn}_i d_i$; $s \leftarrow \sigma(i)$; $k \leftarrow \kappa(i)$; $S_i^{\min} \leftarrow \text{risk}_i \text{crit}_i$
- 2: $\mathcal{A} \leftarrow \emptyset$; $\delta_k \leftarrow (30)$
- 3: **for** each $m \in \mathcal{M}$ **do**
- 4: $z \leftarrow \mathcal{P}(T_i, m, \cdot, t)$ evaluated on the mode-level rules ▷ join of (13)
- 5: **if** $z = \text{prohibit}$ **then**
- 6: **continue**
- 7: **end if**
- 8: $\mathcal{A}_{i,m} \leftarrow$ feasible rosters for m from available $w \in \mathcal{W}_k$, with ψ_a by (4), $\pi_{a,g}^{\text{dev}}$ by (5)
- 9: **if** $z = \text{restrict}$ **then**
- 10: prune $\mathcal{A}_{i,m}$ to rosters satisfying the firing rules' roster guards
- 11: **end if**
- 12: **if** $z = \text{allow-with-oversight}$ **then**
- 13: append the named reviewer to each roster; add reviewer time to Cost ; raise ϕ_m by the assessed retained-judgment share
- 14: **end if**
- 15: **for** each $a \in \mathcal{A}_{i,m}$ **do**
- 16: Query the ten agents for $Q, S, P, Eq^{\text{srv}}, Tr, \text{Cost}, En, Pr$ and $D^{\text{skill}}, D^{\text{fall}}, D^{\text{acct}}, D^{\text{dep}}, D^{\text{trans}}$
- 17: $CAD_{i,m,a} \leftarrow (6)$; $SCV_{i,m,a} \leftarrow (25)$; $\Delta_s^{\text{res}}(m, a) \leftarrow (28)$
- 18: **if** (m, a) satisfies (29) **then**
- 19: $\widetilde{SCV}_{i,m,a} \leftarrow (27)$; $\mathcal{A} \leftarrow \mathcal{A} \cup \{(m, a)\}$
- 20: **end if**
- 21: **end for**
- 22: **end for**
- 23: **if** $\mathcal{A} = \emptyset$ **then**
- 24: $(m, a)_i^* \leftarrow$ Zone Z4 human-reserved roster; record the empty-set event; update $L(t)$; **return**
- 25: **end if**
- 26: $(m, a)_i^* \leftarrow \arg \max_{(m,a) \in \mathcal{A}} \widetilde{SCV}_{i,m,a}$ ▷ Equation (31)
- 27: Report $w_9^*(i)$ by Proposition 3 to the allocation record
- 28: Execute; observe outcomes; $\Lambda_k \leftarrow \Lambda_k + \ell_i \phi_m^* \psi_a^*$
- 29: For each $w \in a^*$: $c_w \leftarrow c_w + \ell_i \phi_m^* \omega_{a^*,w}$; re-evaluate $\text{stage}_k(w, t)$ by (2)
- 30: Update $L(t+1)$, accumulated CAD, $\Pi_{k,g}$, Δ_g , U_r^{used} ; emit to the feedback bus via (32)

Per task, Algorithm 1 performs at most seven policy queries and $\sum_m |\mathcal{A}_{i,m}|$ agent estimates, so the online decision is linear in arrivals and the periodic program is the bottleneck. The rule is exact when the duals are exact and the state has not drifted, which is why the rebalance cadence is tunable and why the tracking gap must be measured across cadences rather than assumed.

4.6 Feasibility of the Capability Budget and the Intake Requirement

A capability budget the task stream cannot supply is not demanding but infeasible, and a framework silent on when this happens will report a failure of its own arithmetic as a failure of the city. Proofs of the four propositions are in Appendix S8.

PROPOSITION 1 (Feasibility of the capability budget). Let $\Phi_k(\Delta T) = \sum_{i \in \mathcal{T}_k(\Delta T)} \ell_i$ be the domain's total developmental content in the period, and let

$$\bar{\zeta}_k = \max\{\phi_m \psi_a : (m, a) \text{ admissible for some } i \in \mathcal{T}_k\}$$

be the largest developmental credit any admissible staffed mode delivers. Then equation (16) is satisfiable only if the condition in equation (33) holds:

$$B_k(t) \leq \Phi_k(\Delta T) \bar{\zeta}_k. \quad (33)$$

Where the estimator (17) is attained by its second argument, this is equivalent to a ceiling on attrition given by equation (34):

$$r_k(t) \leq r_k^{crit} = \frac{\Phi_k(\Delta T) \bar{\zeta}_k}{\eta_k N_k(t) h_k}, \quad (34)$$

and symmetrically to ceilings on η_k and h_k and a floor on the task volume.

Condition (33) binds harder than it appears: machine assistance and supervision ratios both reduce $\bar{\zeta}_k$, safety floors remove the high- ϕ_m candidates on precisely the hazardous tasks the budget most wants, and a trainee under a competent lead can take at most $1 - 1/n_a$ of a roster's human work. When it fails, the framework does not silently relax the constraint. It raises an infeasibility flag, clamps B_k to $\Phi_k \bar{\zeta}_k$, publishes the shortfall as a capability deficit, and reports the four restoring actions: relax the supervision ratio where risk permits, procure less autonomous tooling, expand the supervised task stream, or lower η_k as a recorded decision to let the domain shrink. The framework also needs a hiring lever, which the next proposition supplies.

PROPOSITION 2 (Capability intake requirement). Let Θ_k be the clock hours a developing practitioner works on domain tasks per budget period, $\bar{\phi}_k$ the mean human developmental share of the staffed modes the domain selects, and $\bar{\omega}_k$ the mean work share a developing practitioner holds in those rosters. Then the expected time for one developing practitioner to reach competence is given by equation (35):

$$\tau_k = \frac{h_k^{raw}}{\Theta_k} \cdot \frac{1}{\bar{\phi}_k \bar{\omega}_k}, \quad (35)$$

and in steady state the domain must carry the intake cohort defined in equation (36):

$$\hat{n}_k = \eta_k N_k(t) r_k(t) \tau_k \quad (36)$$

developing practitioners for (16) to be met.

Equation (35) factorizes formation time into the solo-practice time h_k^{raw}/Θ_k that certification assumes and a dilution factor $1/(\bar{\phi}_k \bar{\omega}_k)$ imposed by machine assistance and shared supervision. A city cannot see the dilution factor without this framework; Section 6.6 evaluates it at 3.38, turning a 1.2-year formation into a 4.05-year one.

4.7 What the Price Does and What the Constraint Does

A framework named after a priced construct owes an account of how much of its behavior the price produces.

PROPOSITION 3 (Price and constraint separation). For task i with admissible set $\mathcal{A}(i, t)$, let $\Xi(w_9)$ denote the arg max of (25) with the debt weight set to w_9 and $w_1, \dots, w_8$ held in fixed ratio, which is without loss of generality because the arg max is invariant to positive rescaling of the objective. Define the threshold $w_9^*(i)$ in equation (37):

$$w_9^*(i) = \inf\{w_9 \geq 0 : \Xi(w_9) \neq \Xi(0)\}, \quad (37)$$

with $w_g^* = \infty$ when no finite weight changes the selection. Then for every $w_g < w_g^*(i)$ the selected staffed mode is the one an objective that ignores Civic Automation Debt entirely would select, and any difference in the realized allocation is attributable to the constraints of (26) rather than to the price.

We report w_g^* with every allocation. A domain with $w_g \ll w_g^*$ is a finding rather than an embarrassment: the objective describes the liability while a hard constraint does the work, the division of labor the framework intends. Section 6.2 computes $w_g^* = 0.313$ for the worked case against a default of 0.22.

PROPOSITION 4 (Between-rebalance violation bound). *Suppose the online rule of (31) runs on a window of n arrivals with the admissibility test (29) evaluated on the post-execution state, and let $\bar{\delta}_g = \max_{i,m,a} \delta_g^{disp}(i, m, a)$ and $\bar{u}_r = \max_{i,m,a} u_{i,m,a,r}$. Then at the end of the window the bounds in equation (38) hold:*

$$\Delta_g \leq \tau_g + \bar{\delta}_g \quad \forall g, \quad \sum_i \sum_{m,a} x_{i,m,a} u_{i,m,a,r} \leq U_r \quad \forall r, \quad (38)$$

independently of n , and the resource bound is exact.

Proposition 4 lets the just-transition and resource constraints be enforced online. One task of overshoot is a weak guarantee, but the earlier formulation, which omitted both constraints from the online rule, had none.

4.8 Adaptive Oversight Zones

Task classes are assigned to the four oversight zones of Table 3, which also fixes the staffed modes each zone admits.

Table 3. Adaptive Oversight Zones and the staffed modes each admits. Zone membership is evidence-conditioned by the predicates of Table 4, not a fixed sector label. ϕ^{Z3} defaults to 0.5.

Zone	Allocation principle	Admitted staffed modes
Z1 Autonomous	High-confidence, low-rights-impact tasks may be performed autonomously.	All $m \in \mathcal{M}$; any feasible roster including the null roster.
Z2 Augmented	AI or robots perform substantial work while a human remains meaningfully involved.	m with $\phi_m > 0$; any feasible roster.
Z3 Human Authority	AI may advise, but a designated human authority makes or validates the decision.	m with $\phi_m \geq \phi^{Z3}$; roster must name a competent decision-maker of record who may depart from the recommendation.
Z4 Human Reserved	Automation is restricted for reasons of dignity, democratic authority, capability preservation, or unacceptable systemic risk.	$m \in \{H, H+R\}$ with the robot as a tool under direct human control; roster must contain a competent practitioner.

The tiering, which formalizes the human-in-, on-, and out-of-the-loop taxonomy echoed in the EU AI Act European Parliament and Council of the European Union (2024), is old; what is new is its specified coupling to evidence. Table 4 gives the transition predicates. Promotion toward Z1 requires every condition over a rolling window and an affirmative panel decision, while demotion toward Z4 fires automatically on any single condition, because evidence that automation works should not expand it without a political act, while evidence that it fails should contract it at once.

Table 4. Zone transition predicates. W is a rolling window, four budget quarters by default. Promotion requires all promotion conditions plus an affirmative panel decision; demotion fires on any single demotion condition and is recorded as a versioned policy update. Thresholds are panel-set; the values shown are the defaults listed in Appendix S2 (Table S3).

	Promotion (one zone toward Z1), all required	Demotion (one zone toward Z4), any sufficient
1	No incident of severity 3 or above in W , and incident rate below the class baseline.	One incident of severity 3 or above, or incident rate above $1.5 \times$ the class baseline.
2	Upheld-appeal rate ≤ 0.02 over W .	Upheld-appeal rate > 0.05 over W (the systemic trigger of Section 5.2).
3	Accumulated CAD for the class non-increasing over W .	Accumulated CAD for the class rising for two consecutive quarters.
4	$Res_s^{(n_s)} \geq 1.15 \rho_s$ throughout W .	$Res_s^{(n_s)} < \rho_s$ at any evaluation.
5	$\Lambda_k \geq B_k$ in the last closed budget period.	$\Lambda_k < \bar{B}_k(t)$ for two consecutive quarters, or the infeasibility flag of Proposition 1 is raised.
6	Affirmative Weight Review Panel decision.	Input-distribution drift on the class exceeding a population stability index of 0.25, or a new <i>restrict</i> or <i>prohibit</i> rule entering the twin.

492 A zone change is recorded as a versioned policy update, and restoration after an automatic demotion
 493 always requires a panel decision. Appendix S8.4 collects the notation.

5 GOVERNANCE, CONTESTABILITY, LAWFULNESS, AND DATA PROTECTION

494 5.1 Policy Feedback and Weight Legitimacy

495 Governance is a closed loop: outcomes are evaluated by group, following (32), and feedback modifies
 496 policy parameters through authorized processes. The system operationalizes values set through legitimate
 497 governance rather than deciding them, and fails visibly when those processes have not spoken, leaving an
 498 unset $\theta_{k,g}$ or τ_g flagged rather than quietly defaulted.

499 The weights, budgets, access targets, group coarsening, transition ceilings, zone thresholds, and
 500 contingency parameters are set by a standing *Weight Review Panel* meeting quarterly, to match the
 501 window on which Table 4 evaluates, and after any stress event. It combines a deliberative-polling or
 502 citizen-assembly process for the rights- and distribution-facing parameters with technical review for the
 503 operational ones, following co-creation designs shown to raise trust in municipal AI decisions Phothong
 504 et al. (2026); Nechesov and Ruponen (2024), and records every decision as a versioned update to the Policy
 505 Digital Twin, so any allocation traces to the configuration that authorized it. Three categories of actor must
 506 sit on it. Organized labor has a direct stake in ϕ_m , $B_k(t)$, $\theta_{k,g}$, and τ_g Callari et al. (2024); International
 507 Labour Organization (2015): a panel that sets a displacement ceiling without the workers it will displace is
 508 performing legitimacy rather than exercising it. Workers outside the municipal payroll De Stefano (2016);
 509 Wood et al. (2019) must be represented separately, because $N_k(t)$ does not see them and no incumbent
 510 representative has an incentive to raise their exclusion. And a procurement and legal office must keep
 511 the twin's rule encoding current as law and agreements change, a recurring compliance cost rather than a
 512 one-time setup.

513 Liability follows the zones. In Z3 and Z4 the human decision-maker of record forms a traceable chain;
 514 in Z1 harms trace to the unit that certified the automated component; and in Z2, where a human remains
 515 meaningfully involved, liability is apportioned according to the evidentiary record of which decision steps
 516 each party controlled. Assigning Z2 harms wholly to the vendor, as an earlier version did, contradicts the
 517 human-oversight logic of the EU AI Act European Parliament and Council of the European Union (2024).

5.2 The Contestability Procedure

Contestability in public-sector algorithms must be a first-class design requirement rather than an audit afterthought Moon and Guha (2026); Alfrink et al. (2023), and a system reporting an appeal-resolution rate without specifying an appeal procedure is measuring something it has not built. We specify four elements; Appendix S1 states each in full, and Table S1 is the operative specification of the time limits. *Standing* extends to affected residents, to any worker whose allocation, developmental access, or displacement is at issue, and to their representatives, independently of contract status, with the municipal ombudsman able to act on its own motion where a pattern rather than a single decision is at issue — the systemic-challenge route missing from most deployed systems Vaccaro et al. (2019). *Windows* are twenty working days to lodge and twenty to resolve, halved for resolution and with the outcome suspended where a decision is rights-affecting, and longer for worker appeals on access or displacement, because a worker may not observe the pattern affecting them until a budget period closes. An *evidentiary record* written by every allocation holds the task profile, the roster set before and after policy filtering, each candidate's term vector and *CAD* decomposition, the binding constraint, the duals in force, the flip threshold w_0^* , the configuration version, and the accountable human; it is disclosable to an appellant with standing subject only to third-party redaction. This is technological due process Citron (2008) made operational, and recording the configuration version is what makes an appeal against a *policy* possible, without which a right to human intervention is empty Almada (2019). The *remedy* is re-decision under Zone Z3 by a designated human who may depart from the recommendation on the record, coupled to a systemic trigger: an upheld-appeal rate above 0.05 over the rolling window demotes the class one zone by predicate 2 of Table 4 and refers it to the panel.

The evidentiary record and appeals procedure are separable from the optimizer, and are the framework's most immediately deployable component: a municipality could run them on its existing allocation practice and gain most of the accountability benefit before any optimization is built.

5.3 Lawfulness of the Capability Access Constraint

A mechanism that allocates scarce developmental work against a composition target defined over gender or district is making a decision that equality law regulates, and a paper that invokes the EU AI Act owes the same scrutiny to its own central mechanism. We give the analysis for EU law, which is the most developed, and state what transfers.

Directive 2000/78/EC Council of the European Union (2000) and Directive 2006/54/EC European Parliament and Council of the European Union (2006) prohibit direct discrimination in access to employment, vocational training, and working conditions, and both contain a positive-action provision permitting measures to prevent or compensate for disadvantages linked to a protected ground. The Court of Justice has drawn the boundary in a line of cases beginning with *Kalanke* (C-450/93), which held unlawful a rule giving automatic and unconditional priority to women where candidates were equally qualified, and *Marschall* (C-409/95), which upheld a materially similar rule that carried a saving clause requiring individual assessment. *Abrahamsson* (C-407/98) confirmed that preference for a candidate who is merely sufficiently rather than equally qualified exceeds the exemption. *Badeck* (C-158/97) is the closest authority to our case, because it concerned the reservation of *training places* rather than posts, and the Court accepted such a measure where it did not deprive the other group of the opportunity to train elsewhere. Four design features of the Capability Access Constraint are chosen against that boundary rather than assumed past it.

1. *The constraint binds an aggregate, not an assignment.* Equation (20) constrains the share of a domain's protected practice hours over a full budget period. No individual task is assigned to a practitioner because of a protected characteristic, and no practitioner is excluded from a task by the constraint.
2. *The online channel is a tie-break, not a priority rule.* Between rebalances the target reaches the decision only through the dual $\nu_{k,g}$ in (27), which adds a term to a score that operational terms can outvote. Where two rosters differ materially on merit the merit term dominates, which is the *Marschall* saving clause expressed numerically rather than in prose.
3. *The constraint is elastic.* The slack $\varsigma_{k,g}$ in (26f) means an unattainable target produces a published shortfall, not a forced assignment, so a city can never be placed where the only feasible solution requires an unlawful allocation.

4. *Targets on non-protected grounds carry no such exposure.* The framework's most consequential distributional concern, the exclusion of contracted and platform-mediated practitioners, is a target on *contract status*, which is not a protected ground under either directive; nor is career stage. A municipality unwilling to set a gender or district target can still set the ones that address the framework's own worst failure mode.

Two limits remain, and we state them rather than resolving them. Whether a given $\theta_{k,g}$ is proportionate is a fact-specific judgment on the evidence of under-representation in that domain, which a municipality's legal office must make and which no framework can make for it. And outside the EU the analysis differs: some jurisdictions permit outcome targets that EU law does not, and others prohibit measures EU law permits. What the framework contributes is that the target, its provenance, its tolerance band, and its realized shortfall are versioned, published, and disclosable under Section 5.2, so the proportionality judgment can be made on evidence and challenged on the record. A composition-blind budget makes the same distributional choice invisibly and cannot be challenged at all.

5.4 Data Protection and the Civic Capability Ledger

The ledger of (10) holds per-practitioner practice totals indexed to worker groups defined over gender, contract status, and district, retained across a ten-year horizon and used to allocate work. Under the GDPR European Parliament and Council of the European Union (2016) that is high-risk processing — systematic monitoring of employees that evaluates personal aspects to make decisions with significant effects — so a data protection impact assessment under Article 35 is mandatory before deployment. The design assumes four measures, which Appendix S9 sets out: minimization by aggregation, since the allocator needs only the boolean career-stage test and aggregate shares; separation of protected attributes from the optimizer, which receives only the roster's aggregate group share; small-cell suppression through the $n^{\min}$ rule; and the Article 22 safeguards supplied by the contestability procedure, whose Z3 re-decision remedy is the required human intervention in operational form. Zone assignment is therefore a data-protection mechanism as well as a safety one: a city that places worker-affecting task classes in Z1 should expect to defend that under Article 22 and not only under the AI Act. None of this makes the ledger unproblematic. It makes it a processing operation whose risks are documented, bounded, and challengeable, which is the most a design can offer and less than a worker might reasonably want.

6 A WORKED ALLOCATION: MUNICIPAL BRIDGE INSPECTION

This section carries one allocation through the framework end to end. Every number in Sections 6.2 through 6.9 is exact deterministic arithmetic on the stated inputs and requires no simulation. The inputs are illustrative expert estimates rather than measurements, which is the limitation Section 6.8 addresses by mapping the region of input space over which each conclusion survives.

6.1 The Task, the Domain, and the Candidate Set

T_i is one cycle of fatigue-crack assessment on a steel girder span, with profile $cog = 0.80$, $phy = 0.60$, $emp = 0.10$, $risk = 0.70$, $auth = 0.90$, $priv = 0.20$, $urg = 0.40$, $learn = 0.80$, $crit = 0.90$ and $d_i = 10$ hours, so by (9) its developmental content is $\ell_i = 0.80 \times 10 = 8$ qualified-practice hours. A structural determination of this kind carries a statutory sign-off requirement, so the Policy Digital Twin returns *prohibit* for every mode lacking a licensed human under the encoding of Section 3.4, leaving $\mathcal{M}' = \{H, H+A, H+R, H+A+R\}$.

The safety floor $S_i^{\min} = risk_i crit_i = 0.70 \times 0.90 = 0.63$ removes unaided human inspection ($S = 0.55$) and human plus AI ($S = 0.60$), leaving $\{H+R, H+A+R\}$; earlier presentations left the floor unset and sent a fifth of fracture-critical assessments to unaided teams. Each surviving mode admits two rosters: a_0 , a single competent licensed inspector ($\psi_{a_0} = 0$), and a_1 , a competent lead with one developing inspector on an equal split ($\psi_{a_1} = 0.5$). Because $risk_i = 0.70$, the roster guard allows one developing practitioner per lead, capping ζ_k at $\phi_{H+R} \times 0.5 = 0.35$.

The domain is structural inspection of bridges and major structures: a portfolio on a routine cycle plus fracture-critical and in-depth inspections, generating $n_k = 2,100$ inspection events per year of mean

duration 10 hours and mean learning value 0.80, so $\Phi_k = 2100 \times 8 = 16,800$ qualified-practice hours per year. The workforce is $N_k = 24$ competent practitioners with annual attrition $r_k = 0.12$.

6.2 Pricing the Staffed Modes

Table 5 gives the agents' normalized estimates for the eight staffed modes, the credited developmental share $\phi_m\psi_a$, and the five debt components; applying the default debt weights to (6) and the default objective weights to (25) gives the last two rows.

Table 5. Worked bridge-inspection allocation. Normalized term valuations for the eight staffed modes, resulting Civic Automation Debt from Equation (6), and Sustainable Civic Value from Equation (25). Roster a_0 is a competent lead alone; roster a_1 adds one developing inspector at an equal work split. Shading marks the four staffed modes that survive the safety floor $S_i^{min} = 0.63$. Only D^{skill} depends on the roster; the other four debt components are mode-level.

Term	H		H + A		H + R		H + A + R	
	a_0	a_1	a_0	a_1	a_0	a_1	a_0	a_1
<i>Flow terms, measured on this execution window</i>								
Q service quality	0.72	0.69	0.86	0.83	0.80	0.77	0.91	0.88
S safety	0.55	0.53	0.60	0.58	0.88	0.86	0.90	0.88
P productivity	0.40	0.35	0.62	0.57	0.70	0.65	0.88	0.83
Eq^{srv} service equity	0.70	0.70	0.72	0.72	0.70	0.70	0.74	0.74
Tr trust	0.80	0.78	0.74	0.72	0.72	0.70	0.68	0.66
$Cost$ operating cost	0.85	0.95	0.62	0.72	0.58	0.68	0.50	0.60
En energy	0.20	0.20	0.28	0.28	0.55	0.55	0.60	0.60
Pr privacy risk	0.10	0.10	0.25	0.25	0.30	0.30	0.38	0.38
<i>Stock-change terms, settled later</i>								
ϕ_m human share	1.00	1.00	0.75	0.75	0.70	0.70	0.55	0.55
ψ_a developmental eligibility	0.00	0.50	0.00	0.50	0.00	0.50	0.00	0.50
$\phi_m\psi_a$ credited share	0.000	0.500	0.000	0.375	0.000	0.350	0.000	0.275
$D^{skill} = 1 - \phi_m\psi_a$	1.00	0.50	1.00	0.625	1.00	0.650	1.00	0.725
D^{fall} fallback erosion	0.00	0.00	0.15	0.15	0.25	0.25	0.35	0.35
D^{acct} accountability dilution	0.00	0.00	0.07	0.07	0.05	0.05	0.13	0.13
D^{dep} vendor dependency	0.00	0.00	0.30	0.30	0.35	0.35	0.50	0.50
D^{trans} transition burden	0.00	0.00	0.10	0.10	0.20	0.20	0.30	0.30
$CAD_{i,m,a}$ from (6)	0.2800	0.1400	0.3816	0.2766	0.4260	0.3280	0.5004	0.4234
$SCV_{i,m,a}$ from (25)	0.2324	0.2391	0.2783	0.2773	0.3108	0.3082	0.3426	0.3355
Survives $S_i^{min} = 0.63$?	no	no	no	no	yes	yes	yes	yes

Read alone, Table 5 favors a fully hybrid team staffed by a lead alone: $H+A+R/a_0$ has the best quality, safety, and productivity, the lowest cost, and the highest SCV at 0.3426, despite the largest debt of the eight at 0.5004. Pricing debt does not by itself change this. With the other weights in fixed ratio, the arg max first changes at the threshold of Proposition 3,

$$w_9^* = 0.313, \quad (39)$$

where $H+A+R/a_1$ overtakes $H+A+R/a_0$, against a default $w_9 = 0.22$. The price narrows the gap between the two rosters from 0.0241 to 0.0072 objective units, closing 70% of it; on these inputs the reversal is produced by the hard constraint, not the price.

6.3 The Capability Constraint Binds

Certification at inspector grade requires $h_k^{raw} = 1,800$ supervised clock hours, so (18) gives $h_k = 0.80 \times 1800 = 1,440$; we target independent competence rather than statutory team-leader authority, which is a separate legal threshold, and note that h_k is the parameter to which the example is most sensitive. With

633 $\eta_k = 1.2$ and $B_k^{min} = 2,000$ hours, Equation (17) yields

$$B_k = \max(2000, 1.2 \times 24 \times 0.12 \times 1440) = 4,976.64 \text{ qualified-practice hours per year.} \quad (40)$$

634 The budget requires a mean credited share of at least $4976.64/16800 = 0.2962$, and Proposition 1 holds
 635 with little room, at $\Phi_k \zeta_k = 5,880$. The unconstrained allocation delivers nothing: every inspection sent
 636 to $H+A+R/a_0$ credits zero, where earlier presentations credited an incumbent working alone with a
 637 1,496-hour delivery. A domain staffed entirely by its incumbents forms no successors.

638 6.4 Solving the Program and Reading Its Price

639 Restricted to this domain and its single binding constraint, program (26) has an optimum with at most
 640 two active staffed modes, because one coupling constraint plus the simplex constraint (26b) admits at most
 641 two nonzero variables at an LP vertex. Table 6 enumerates every admissible pair reaching the required
 642 share.

Table 6. All two-mode mixes of admissible staffed modes reaching the required mean credited share $\phi_m \psi_a = 0.2962$. The enumeration is complete: no pair excluding roster a_1 can reach a positive credited share at all, and a single coupling constraint admits at most two active modes at an LP vertex. Bold marks the optimum.

Mix	Shares of tasks	Mean $\phi_m \psi_a$	Mean <i>SCV</i>
$H+R/a_1$ with $H+A+R/a_1$	28.30% / 71.70%	0.2962	0.3277
$H+R/a_1$ with $H+A+R/a_0$	84.64% / 15.36%	0.2962	0.3135
$H+R/a_1$ with $H+R/a_0$	84.64% / 15.36%	0.2962	0.3086

643 The optimum sends 28.30% of inspections to robot-assisted human teams and 71.70% to fully hybrid
 644 teams, and staffs a developing inspector on *every* one of them. The capability constraint's dual follows
 645 from the tie between the two active staffed modes, equation (41):

$$0.308240 + 8 \lambda_k(0.350) = 0.335452 + 8 \lambda_k(0.275) \implies \lambda_k = \frac{0.027212}{0.6} = 0.045353, \quad (41)$$

646 with the two *SCV* values carried at six decimals because their difference is the whole numerator. One
 647 additional qualified-practice hour therefore costs 0.0454 objective units at the margin. This is the marginal
 648 *opportunity cost* of tightening the constraint, not the value of an hour of practice, which the framework
 649 neither estimates nor needs; earlier presentations inverted the reading. Whether 0.0454 units per hour is
 650 worth paying is the panel's decision. Substituting into (27) reproduces the ranking without solving the
 651 program, equation (42):

$$\begin{aligned} \widetilde{SCV}_{H+R/a_1} &= \widetilde{SCV}_{H+A+R/a_1} = 0.435229 \\ &> \widetilde{SCV}_{H+A+R/a_0} = 0.3426 \\ &> \widetilde{SCV}_{H+R/a_0} = 0.3108. \end{aligned} \quad (42)$$

652 The first equality holds by construction. The capability price reverses the winner: the same objective,
 653 estimates, and weights now select a trainee-staffed team and move over a quarter of inspections to robot-
 654 assisted human work, so the reversal falls first on *who is on the team* and only second on *which machines*
 655 *are used*. Integrally, 594 tasks in $H+R/a_1$ deliver 4,976.40 hours and miss the budget by 0.24, while 595
 656 deliver 4,977.00, so the integer optimum's mean *SCV* of 0.327742 sits 7.8×10^{-6} below the LP value of
 657 0.327750.

6.5 What Preservation Costs and What It Buys

Unconstrained allocation yields a mean SCV of 0.3426 per task, or 719.5 objective units a year over 2,100 tasks; the constrained optimum yields 0.3277, or 688.3 units. The constraint costs 31.2 units a year, a reduction of 4.34%, and over the 4,976.64 hours it purchases the average price is 0.0063 units per hour, below the marginal 0.0454 as convexity requires. Table 7 decomposes which terms absorbed that cost, because an aggregate figure hides whether a city bought its pipeline with money, with quality, or with safety.

Table 7. Term-by-term decomposition of the cost of preservation in the worked domain: the unconstrained allocation (all tasks to $H+A+R/a_0$) against the constrained optimum. Cost, energy, and privacy risk are penalties in Equation (25), so a fall is an improvement for those three rows.

Term	Unconstrained	Constrained optimum	Change
Q service quality	0.9100	0.8489	−6.72%
S safety	0.9000	0.8743	−2.85%
P productivity	0.8800	0.7791	−11.47%
Eq^{srv} service equity	0.7400	0.7287	−1.53%
Tr trust	0.6800	0.6713	−1.28%
$Cost$ operating cost	0.5000	0.6226	+24.53%
En energy	0.6000	0.5858	−2.36%
Pr privacy risk	0.3800	0.3574	−5.96%
CAD civic automation debt	0.5004	0.3964	−20.78%
Credited share $\phi_m\psi_a$	0.0000	0.2962	from zero
SCV	0.3426	0.3277	−4.34%

The city pays mainly in operating cost, up 24.5% because every inspection carries a second person, and productivity, down 11.5%, with smaller losses of 6.7% in quality and 2.9% in safety, both above the floor. Preservation costing 4.34% of objective value and preservation costing a quarter of the inspection budget are the same fact, but only the second is the one a finance committee will contest.

What the city buys is a change of sign. Under the constraint the domain generates $4976.64/1440 = 3.456$ competent-equivalents a year against attrition of $24 \times 0.12 = 2.88$, a net gain of 0.576; without it, it forms no one and loses 2.88 a year. The identity $B_k/h_k = \eta_k N_k r_k$ makes this replenishment rate independent of h_k , even though feasibility is not.

6.6 Pipeline Arithmetic and the Intake Requirement

The optimum staffs a developing inspector on all 2,100 inspections, consuming 21,000 trainee clock hours a year, each of which accrues only $4976.64/21000 = 0.237$ qualified-practice hours, because practice is diluted by machine assistance ($\bar{\phi}_k = 0.5925$ at the optimum) and by sharing the human work with a lead ($\bar{\omega}_k = 0.5$). At $\Theta_k = 1,500$ domain clock hours per trainee-year, Proposition 2 gives the formation time in equation (43):

$$\tau_k = \frac{1800}{1500} \cdot \frac{1}{0.5925 \times 0.5} = 1.20 \times 3.376 = 4.05 \text{ years}, \quad (43)$$

and the intake requirement in equation (44):

$$\hat{n}_k = 1.2 \times 24 \times 0.12 \times 4.05 = 14.0 \text{ developing practitioners}. \quad (44)$$

Both routes agree: 21,000 trainee clock hours at 1,500 per trainee-year is exactly 14.0 posts, and Little's law at 3.456 completions a year and 4.05 years' residence gives the same, an identity that lets a city compute its intake from its task stream without modeling its pipeline.

The dilution factor is the finding. Certification assumes 1,800 supervised hours, 1.2 years of solo practice; under the optimal machine-assisted supervised mix the same practitioner takes 4.05 years, 3.38 times longer, and a plan built on the certification figure would size intake at four trainees where fourteen are needed. No

headcount plan surfaces this, because the dilution lives in $\bar{\phi}_k$ and $\bar{\omega}_k$, properties of the *allocation* rather than the workforce. A city that automates more aggressively forms each practitioner more slowly as well as forming fewer, and will not see the second effect until the first has happened.

6.7 Who Receives the Protected Work

The 4,976.64 protected hours must go to someone, and (16) does not say to whom. Suppose the 24 incumbents are 21 men and three women, 87.5% to 12.5%, while the qualified local labor pool stands at 68% to 32%. A composition-blind budget recruits through the channels that produced the incumbents and distributes protected practice in proportion, sending $0.125 \times 4976.64 = 622$ hours a year to the under-represented group, or $622/1440 = 0.432$ competent-equivalents against a domain-wide 3.456. The cadre's composition is not merely preserved but re-founded, one replacement cycle at a time.

Constraint (20) intervenes on the distribution rather than the size. Setting $\theta_{k,g}$ to the labor-pool composition of 0.32 with $\varepsilon_k = 0.05$ requires $\Pi_{k,g} \geq 0.27$, or at least 1,344 hours a year, equal to 0.933 competent-equivalents: 2.16 times the composition-blind rate, achieved without changing B_k , the weights, or any estimate, and meaning 3.8 of the 14 developing posts of (44) rather than 1.8. The point is not the multiplier, which follows from illustrative composition figures, but that who wins is set by a published parameter a panel chose and can be asked to defend, subject to Section 5.3. Table 8 gives the whole distribution both ways; because protected practice is scarce at the optimum the access constraint binds with equality, so every cell is a determinate consequence of B_k and the panel's choice of $\theta_{k,g}$ and ε_k .

Table 8. Distribution of the $B_k = 4,976.64$ protected practice hours in structural inspection, derived by exact arithmetic. “Composition-blind” distributes protected practice in proportion to the incumbent composition; “Under (20)” applies the binding floor $\Pi_{k,g} \geq \theta_{k,g} - \varepsilon_k$ with $\theta_{k,g} = 0.32$ and $\varepsilon_k = 0.05$. Competent-equivalents are hours divided by $h_k = 1,440$. Groups are the marginal partition on gender and contract status, with the intersection suppressed under the n^{min} rule. The contracted row is not a result and not a modelling choice: its share is structurally zero for any $\theta_{k,g}$, because non-payroll practitioners are absent from N_k and ineligible as roster members under (19).

Group	Incumbent	$\theta_{k,g}$	Composition-blind			Under (20)		
			$\Pi_{k,g}$	h/yr	c.eq.	$\Pi_{k,g}$	h/yr	c.eq.
Men, permanent	0.875	0.680	0.875	4,354.56	3.024	0.730	3,632.95	2.523
Women, permanent	0.125	0.320	0.125	622.08	0.432	0.270	1,343.69	0.933
Contracted, any gender	—	unset	0.000	0.00	0.000	0.000	0.00	0.000
Total on payroll	1.000	—	1.000	4,976.64	3.456	1.000	4,976.64	3.456

One group is invisible in these numbers, and the third row of Table 8 is the only honest way to show it. If 40% of this domain's inspection hours were delivered by contracted rope-access technicians off the municipal payroll, they would not appear in N_k , would not be eligible roster members, and would receive a structurally zero share whatever $\theta_{k,g}$ is, with their displacement unrecorded. That is a property of the data substrate rather than of the allocation, so no setting of the access constraint reaches it; Sections 7.4 and 7.6 treat it as the framework's most consequential distributional limitation.

6.8 Sensitivity and the Feasibility Frontier

Every conclusion above rests on expert estimates, so Table 9 sweeps the six load-bearing parameters one at a time, reporting the optimum, the dual, the cost of preservation, and whether Proposition 1 holds, all by exact arithmetic.

Four things follow, and two of them are uncomfortable. First, the qualitative staffing result holds across the frontier and its magnitude does not. No lead-only roster delivers any credit and the budget is strictly positive, so some developmental staffing occurs at every feasible parameter value we swept; but across the eight feasible rows labeled “no reversal” the constrained optimum retains the lead-only roster on between 1.3% and 46.1% of tasks, and the dual falls from 0.0454 to between 0.0005 and 0.0033. A binding capability budget always places developing practitioners on some of the work, not on essentially all of

Table 9. One-at-a-time sensitivity of the worked allocation. Baseline row in bold. “Infeasible” means Proposition 1 fails, so the budget exceeds what the task stream can supply and the escalation of Section 4.6 is triggered. “No reversal” means the constrained optimum retains the unconstrained lead-only roster on some tasks, so the capability price does not change the staffing decision.

Parameter	Value	Optimum	λ_k	Cost	Status
h_k^{raw} (clock h)	1200	28.2% $H+A+R/a_0$ + 71.8% $H+A+R/a_1$	0.0033	1.50%	no reversal
	1500	10.2% $H+A+R/a_0$ + 89.8% $H+A+R/a_1$	0.0033	1.88%	no reversal
	1800	28.3% $H+R/a_1$ + 71.7% $H+A+R/a_1$	0.0454	4.34%	baseline
	2127	boundary of Proposition 1	—	—	critical
	2400	—	—	—	infeasible
	3000	—	—	—	infeasible
r_k (per year)	0.06	46.1% $H+A+R/a_0$ + 53.9% $H+A+R/a_1$	0.0033	1.13%	no reversal
	0.09	19.2% $H+A+R/a_0$ + 80.8% $H+A+R/a_1$	0.0033	1.69%	no reversal
	0.12	28.3% $H+R/a_1$ + 71.7% $H+A+R/a_1$	0.0454	4.34%	baseline
	0.14	94.1% $H+R/a_1$ + 5.9% $H+A+R/a_1$	0.0454	9.57%	near-critical
	0.1418	boundary of Proposition 1	—	—	critical
η_k	0.8	28.2% $H+A+R/a_0$ + 71.8% $H+A+R/a_1$	0.0033	1.50%	no reversal
	1.2	28.3% $H+R/a_1$ + 71.7% $H+A+R/a_1$	0.0454	4.34%	baseline
	1.4	94.1% $H+R/a_1$ + 5.9% $H+A+R/a_1$	0.0454	9.57%	near-critical
	1.6	—	—	—	infeasible
ϕ_{H+R}	0.55	—	—	—	infeasible
	0.60	84.9% $H+R/a_1$ + 15.1% $H+A+R/a_1$	0.1515	9.60%	reversal
	0.70	28.3% $H+R/a_1$ + 71.7% $H+A+R/a_1$	0.0454	4.34%	baseline
	0.85	14.2% $H+R/a_1$ + 85.8% $H+A+R/a_1$	0.0188	3.02%	reversal
ϕ_{H+A+R}	0.45	57.0% $H+R/a_1$ + 43.0% $H+A+R/a_1$	0.0241	7.00%	reversal
	0.55	28.3% $H+R/a_1$ + 71.7% $H+A+R/a_1$	0.0454	4.34%	baseline
	0.60	1.3% $H+A+R/a_0$ + 98.7% $H+A+R/a_1$	0.0023	1.62%	no reversal
	0.70	15.4% $H+A+R/a_0$ + 84.6% $H+A+R/a_1$	0.0009	0.63%	no reversal
ψ_{a_1} (supervision)	0.40	—	—	—	infeasible
	0.4232	boundary of Proposition 1	—	—	critical
	0.50	28.3% $H+R/a_1$ + 71.7% $H+A+R/a_1$	0.0454	4.34%	baseline
	0.67	19.6% $H+A+R/a_0$ + 80.4% $H+A+R/a_1$	0.0005	0.33%	no reversal

it; the latter is a property of the baseline parameterization. Second, the *mode* reversal is not robust: it requires $\phi_{H+A+R} \leq 0.55$ and $\psi_{a_1} \leq 0.5$, and raising the fully hybrid mode’s developmental share to 0.60, or allowing two developing practitioners per lead, lets that mode nearly satisfy the budget alone, collapsing the dual by an order of magnitude. That is correct behavior, since a mode that develops people should not be penalized, but the headline mode shift is a statement about one estimate of ϕ_{H+A+R} . Third, feasibility is fragile. The baseline sits at 85% of the critical value on all three axes Proposition 1 bounds, so raising h_k^{raw} to 2,400 hours, well within a stricter licensing regime, or attrition to 0.16, which a retirement wave produces easily, makes the budget unattainable: the domain has roughly 15% headroom on three separate axes. Fourth, the cost of preservation ranges from 0.33% to 9.60% across the swept region, and wherever this article quotes 4.34%, Table 9 is the range that should accompany it.

6.9 Two Parameter Shocks: Stability and Infeasibility

Two displacements are worth carrying through, because they are the two regimes Proposition 1 separates and a city will meet both. Both are exact arithmetic on the model, not simulation output; Appendix S10 gives every step.

Shock one: a retirement wave inside the feasible region. Raising attrition from 0.12 to 0.14, inside the critical rate of 0.1418, expands the budget to 5,806.08 hours and the required credited share from 0.2962 to 0.3456. The same pair of modes still brackets the requirement, so λ_k stays at 0.0454 and only the mix moves: robot-assisted human work rises from 28.3% to 94.1%, the cost of preservation from 4.34% to 9.56%, and the domain operates at 98.7% of its ceiling. This is the stabilizing direction the capability feedback is designed to have. The arithmetic does *not* establish that an implementation updating duals at a

finite rebalance cadence converges to this point rather than oscillating around it; Proposition 4 bounds drift between rebalances by one task, not the trajectory, and no result in this article does.

Shock two: a retirement wave outside it. At attrition 0.25 the budget becomes 10,368 hours against the ceiling of 5,880, a deficit of 4,488 hours or 3.12 competent-equivalents a year that *no* allocation of this task stream can form. Of the four restoration options of Section 4.6, permitting two developing practitioners per lead leaves a residual deficit of 2,528 hours; expanding the supervised stream to 3,703 inspections a year closes it; lowering η_k to 0.68 closes it by deciding not to replace the full cadre, a structural contraction that should be recorded as one; and accepting the deficit logs the shortfall against a named decision-maker. The lesson concerns task structure rather than recruitment. The binding quantity is $\Phi_k \bar{\zeta}_k$, the practice the task stream can carry: a municipality facing this shock could recruit thirty trainees and still fail, because the stream supports at most 14 supervised trainee posts at the prevailing automation level. When a capability budget becomes infeasible, the remedy lies in the allocation or in the scope of the cadre the city intends to replace, not in the intake alone.

7 DISCUSSION

7.1 An Analytic Model of Debt Accumulation

The debt components of (6) are per-task quantities; what a city experiences is their accumulation, and its shape distinguishes a strategy that replenishes a stock from one that does not. Let $\pi_j(t) \in [0, 1]$ be the fraction of component j 's required replenishment delivered at time t , let debt accrue in proportion to the unmet fraction, and let each component approach its own asymptote with its own time constant, equation (45):

$$\dot{C}_j(t) = c_j [1 - \pi_j(t)], \quad \pi_j(t) = \pi_{j,\infty} (1 - e^{-t/\tau_j}). \quad (45)$$

Integrating from $\pi_j(0) = 0$ and weighting by the debt weights of (6) gives the total debt trajectory in equation (46):

$$C(t) = \sum_j w_j c_j \left[(1 - \pi_{j,\infty}) t + \pi_{j,\infty} \tau_j (1 - e^{-t/\tau_j}) \right], \quad w_j \in \{\alpha, \beta, \gamma, \delta, \epsilon\}. \quad (46)$$

Earlier presentations modeled one scalar $\pi(t)$, which is D^{skill} alone, and plotted it as CAD. Modeling all five matters because different mechanisms drive each $\pi_{j,\infty}$: the hard constraint (16) drives skill formation to one, with $\tau_{skill} = 4.05$ years taken from Proposition 2; the reserve (24) drives fallback to one; revisable zones and (21) drive accountability and transition high; and *nothing in the framework drives vendor dependency*, which keeps accruing linearly. Table 10 fixes every parameter and Figure 3 plots (46) at exactly those values.

Table 10. Per-component replenishment parameters for Equation (46), one row per strategy, reported as $\pi_{j,\infty}$ with τ_j in years in parentheses. The accrual rate is $c_j = 10$ index units per year for every component of every strategy, except that Cost/Performance carries $c_{acct} = 14$ and $c_{dep} = 16$. These are the article's illustrative choices, not measured quantities; they are tabulated so that Figure 3 is reproducible by direct substitution rather than asserted. Only $\tau_{skill} = 4.05$ is derived, from Proposition 2. The final two columns are computed from the row, not fitted to the plot.

Strategy	$\pi_{skill,\infty}$ (τ)	$\pi_{fall,\infty}$ (τ)	$\pi_{acct,\infty}$ (τ)	$\pi_{dep,\infty}$ (τ)	$\pi_{trans,\infty}$ (τ)	Residual slope	$C(10)$
Automation-First	0	0	0	0	0	10.00	100.00
Cost/Performance	0	0	0	0	0	11.44	114.40
Capability Matching	0.35 (4.05)	0.30 (2.0)	0	0	0	8.36	88.54
Ergonomics-Aware Role Allocation	0.40 (4.05)	0.45 (2.0)	0	0	0	7.89	85.02
Human-First	0.15 (4.05)	1.00 (0.5)	1.00 (0.5)	1.00 (0.5)	1.00 (0.5)	2.38	28.96
CivicWorkOS	1.00 (4.05)	1.00 (1.0)	0.85 (1.5)	0.10 (2.0)	0.95 (1.0)	1.45	31.51

Three things follow from Figure 3, and one of them is unfavorable to the framework.

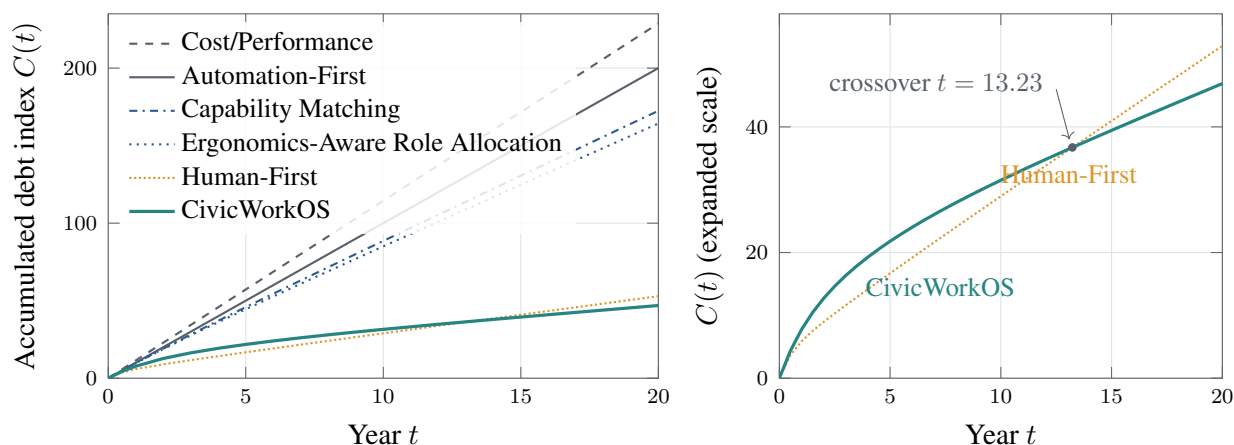


Figure 3. Accumulated Civic Automation Debt under Equation (46) at the per-component parameters of Table 10. Every plotted point is that equation evaluated at those parameters; nothing here is measured or simulated. *Left:* all six strategies over twenty years. Automation-First is normalized to 100 at year ten by setting $c_j = 10$ for every component, and Cost/Performance exceeds it because it carries elevated accountability and dependency accrual. *Right:* the two capability-forming strategies at an expanded scale. CivicWorkOS drives skill formation and fallback to full replenishment through hard constraints, with $\tau_{skill} = 4.05$ years derived from Proposition 2, but has no mechanism for vendor dependency, so its curve does not saturate. Human-First accumulates less debt than CivicWorkOS until year 13.23 and more thereafter, because its residual slope of 2.38 index units per year, almost all of it unformed skill, exceeds CivicWorkOS's 1.45, three-quarters of which is vendor dependency.

768 *No strategy here bounds debt, CivicWorkOS included.* Its saturating components contribute a bounded
 769 17.98 index units, 11.34 of them from skill formation, but the residual accrues at 1.45 units a year without
 770 limit, 1.08 of it vendor dependency. A capability budget converts *skill* debt from unbounded to bounded
 771 and says nothing about the other components except through outvotable weights. Vendor dependency is a
 772 procurement variable the allocation mechanism does not control, and pretending otherwise would repeat
 773 the category error Section 3.6 corrects for reskilling coverage.

774 *Human-First beats CivicWorkOS on this metric for thirteen years.* Keeping humans on everything
 775 replenishes fallback, accountability, dependency, and transition at once, while CivicWorkOS fills its
 776 pipeline over a 4.05-year time constant. The curves cross at $t = 13.23$, both at 36.73 index units, and
 777 by year twenty stand at 46.89 against 52.89, because Human-First never closes its skill gap: maximizing
 778 human share is not forming practitioners. On a ten-year horizon and this index alone a city should prefer
 779 Human-First, and would pay for it in operational terms the index omits, which is why debt is one construct
 780 among thirteen rather than the objective.

781 *The ordering of the curves is not evidence.* It follows analytically from the mechanisms each strategy
 782 enforces, so Figure 3 describes the debt accounting, not cities. Whether $\pi_{skill,\infty}$ can reach one across
 783 heterogeneous task streams without excessive Zone Z4 fallback is what the protocol of Section 7.3 would
 784 answer.

785 7.2 What the Framework Establishes, and What It Does Not

786 Section 6 establishes by arithmetic that the constraint system is computable on a realistic domain, that its
 787 constraints bind, that the price of the binding constraint is legible and publishable, and that the distributional
 788 question has a determinate answer with (20) and none without it; Section 6.8 bounds those results, and
 789 the four propositions establish satisfiability, the response to infeasibility, the price's share of the behavior,
 790 and the online drift bound. None of that establishes that any input estimate is correct. The estimates are
 791 the authors' structured judgment, the elicitation protocol of Appendix S5 has not been run, and sensitivity
 792 analysis is a weaker substitute for measurement.

793 Nor is a property of the construction a finding. Because $-w_9CAD$ sits in the same weighted sum as P
 794 and $-Cost$ under hard capability and resilience constraints, any *SCV*-maximizing strategy sacrifices some

productivity and cost for capability; that is a tautology, and it applies to hypotheses P2, P3, P5, and P9 as much as to the general claim. The substance lies in magnitudes — whether the sacrifice is 4% or 40%, whether the system is satisfiable at municipal scale, whether the price is stable — of which Section 6.8 bounds the first between 0.33% and 9.60% for one domain.

7.3 A Specified Evaluation Protocol

The protocol described in this subsection is specified in full and *has not been executed*. It fixes the evaluation this framework requires in advance, so that it can be reproduced by others and audited against a plan rather than one assembled after outcomes are known, and nothing in this article reports an outcome of it; Appendix S2 gives the complete specification. It is a discrete-event testbed over six municipal sectors and 803,100 tasks a year, run for ten years so that retirement waves and pipeline decay fall inside the horizon, comparing Automation-First, Cost/Performance, Human-First, and CivicWorkOS — all configurations of (25) — with the external baselines of Ranz et al. (2017) and Merlo et al. (2023), under paired Wilcoxon tests with Holm–Bonferroni control over sixty tests and seven stress scenarios, two of which Section 6.9 works out by exact arithmetic.

An evaluation in which four of six comparators are configurations of the proposed objective can show that the mechanism does what it was designed to do; it cannot show that it outperforms an independently designed alternative. The two external baselines are specified for that reason, and the comparison is incomplete until they are run; neither is implemented at the time of writing. The calibration portals also belong to high-income cities with few non-payroll workers, so the evaluation would understate the population the framework fails; an earlier version of this article wrongly called that bias conservative.

The hypotheses are *pre-specified*, not pre-registered, since a manuscript is not a timestamped registry entry; they will be lodged on the Open Science Framework before execution and the identifier cited in whichever article reports the outcome. Their thresholds were also set with the worked example's arithmetic in hand, a weaker form of pre-specification than setting them blind, and we claim no more.

7.4 Who Wins and Who Loses

The distributional question has four answers in this framework, and they are not equally comfortable.

The clearest winners are residents of a city whose critical services retain human fallback: constraint (24) makes an AI-service outage a degradation rather than an interruption. Its shape matters. It reserves capacity against the loss of a contributor, not against aggregate demand, so a reserve sized for an outage is not one sized for a surge.

The second group is workers who receive protected developmental work, provided $\theta_{k,g}$ is deliberately calibrated rather than inherited. A capability budget rations a scarce good, and under a composition-blind budget the losers are precisely the under-represented groups, at 622 hours a year against a possible 1,344 in Section 6.7.

The third group is the one the framework serves worst. Workers outside the municipal payroll — contracted, agency, informal, or platform-mediated De Stefano (2016); Wood et al. (2019) — are absent from $N_k(t)$, ineligible as roster members, and unprotected by (21). Where they perform much of a city's sanitation, inspection, and logistics labor, CivicWorkOS as specified would preserve the permanent workforce while the contracted one absorbed the displacement, and report a satisfied access constraint while doing so; Table 8 prints the structurally zero row rather than hiding it. No parameter setting fixes this, because it follows from a data substrate that presumes formal employment. Extending the ledger to non-payroll practitioners is technically straightforward and institutionally hard.

The fourth answer concerns automation itself, which CivicWorkOS legitimates as much as it constrains: a documented *SCV* computation, a satisfied budget, and a versioned panel decision can justify displacing workers once the price is paid. Whether that is a feature or a hazard depends on who sets the weights, which is why panel composition is a design parameter and why Proposition 3 lets a city see whether its objective or its constraints are deciding. The framework is not neutral between contested outcomes; it makes a political choice explicit and traceable, and should be judged as such.

843 7.5 Threats to Validity

844 *Construct validity.* Qualified-practice hours are a proxy, and hours are not learning; without the retention
 845 functions of the skill-decay literature Arthur et al. (1998); Tatel and Ackerman (2025), the ledger overstates
 846 competence under intermittent practice. *Internal validity.* Most terms are reported by agents with an interest
 847 in their value, and ϕ_m , which enters four channels, comes from the agent whose interest is a larger budget;
 848 cross-validation against observed outcomes is only a partial remedy. *External validity.* One city scale and a
 849 high-income calibration base say nothing about a majority-informal workforce, where the problem is most
 850 severe. *Statistical validity.* The specified design detects standardized effects of $d = 0.76$ or larger, so it
 851 fixes in advance the rule for raising seed counts.

852 7.6 Limitations

- 853 • *No measured inputs.* The estimates and parameters are the authors' own; the elicitation protocol that
 854 would replace them (Appendix S5) has not been run.
- 855 • *Weight legitimacy.* The quarterly panel of Section 5.1 requires deliberative processes a municipality
 856 would have to build, and some parameters move faster than any panel meets.
- 857 • *Linear scalarization.* A weighted sum reaches only the convex hull of the attainable set, so the panel's
 858 authority is bounded by the aggregation function rather than its mandate; Chebyshev or ε -constraint
 859 formulations would remove this at some cost in interpretability.
- 860 • *No guarantee for the online rule.* Beyond Proposition 4 there is no regret bound, no dual-tracking
 861 condition, and no approximation guarantee; the constrained-MDP literature Altman (1999); García and
 862 Fernández (2015) holds the machinery, and connecting (27) to it is the most valuable missing theory.
- 863 • *Unmeasured tractability.* Program (26) is NP-hard in general, and no claim here depends on where it
 864 stops being tractable.
- 865 • *No envy guarantee.* The access constraint is a share constraint, not a fair-division mechanism Bouveret
 866 et al. (2016), so members of one group can receive very different practice.
- 867 • *Privacy and gaming.* The measures of Section 5.4 bound but do not remove the ledger's exposure, and
 868 ϕ_m is the most gameable quantity in the framework.
- 869 • *Machine-checkability and jurisdiction.* Decidable predicates cannot capture the discretionary judgment
 870 that constitutes much of a public service Lipsky (2010), and Section 5.3 covers EU law only.
- 871 • *Formal-employment presumption.* The framework preserves the visible payroll workforce and cannot
 872 see the rest. This is the limitation we would fix first.

8 8 CONCLUSIONS

873 The AI era requires a richer question than whether humans, agents, or robots work better, because the
 874 allocation itself changes what a city will be able to do, and who will be able to do it. CivicWorkOS treats
 875 urban automation as a governed portfolio rather than a sequence of independent efficiency decisions.

876 Its central proposition is that human expertise, fallback competence, accountability, and democratic
 877 control are renewable but depletable civic resources, and that automation consuming them without
 878 replenishment creates Civic Automation Debt. The concern has established lineages in automation
 879 research Bainbridge (1983), debt accounting Cunningham (1992), and personnel scheduling Burke et al.
 880 (2004); Van den Bergh et al. (2013). The contribution is a decision variable that names who performs the
 881 work, a capability constraint that accrues only to the people it is sized to form, an access constraint that
 882 makes the resulting distribution an explicit and contestable choice, and the formal results that say when the
 883 system is satisfiable and what to do when it is not.

884 The worked allocation shows what this buys in an end-to-end municipal case. A capability price of 0.0454
 885 objective units per practice hour reverses the winning staffed mode, costs 4.34% of annual objective value,
 886 and turns a domain that would have formed nobody into one that generates 3.46 competent-equivalents a
 887 year against attrition of 2.88. An access constraint more than doubles the share of that formation reaching
 888 an under-represented group, from 622 to 1,344 hours a year, without changing the budget, the weights,
 889 or any input estimate. And the framework's own arithmetic produces a result no headcount plan surfaces:
 890 machine-assisted supervised practice stretches time to competence by a factor of 3.38, from 1.2 years to

4.05, so a city automating its junior work does not merely form fewer practitioners, it forms each remaining one more slowly, and will not see the second effect until the first has already happened.

Three limits bear restating. The framework bounds skill debt but not vendor-dependency debt, so Figure 3 does not saturate. Its feasibility margin in the worked domain is roughly 15% on three axes, so a stricter certification regime or an older workforce reaches the boundary rather than the optimum. And it cannot see workers who are not on a payroll, the population the exposure literature identifies as most exposed.

Future work should execute the elicitation protocol so the inputs are measured; connect the online rule to the constrained-MDP literature for regret and violation bounds; extend the ledger to contracted and platform-mediated practitioners; let c_w depreciate under intermittent practice; and pilot the evidentiary record and contestability procedure of Section 5.2 against an existing municipal allocation practice, which a city could adopt tomorrow without the optimizer.

ABBREVIATIONS

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DATA AVAILABILITY STATEMENT

No new datasets were created or analyzed in this study. All parameter values, the worked allocation of Section 6, the sensitivity sweep of Section 6.8, the parameter shocks of Section 6.9, and the debt trajectory of Section 7.1 are fully specified in this article and can be reproduced from the reported values by direct arithmetic. The author-provided reference implementation, which recomputes every checkable number this article publishes and reports the agreement check by check, is available at <https://github.com/aliakarma/CivicWorkOS>. Code under `sim/` in that repository runs on clearly labelled synthetic demonstration data; it exercises the software shape of the protocol specified in Section 7.3 and Appendix S2, and it must not be cited as empirical municipal validation. The evaluation protocol of Section 7.3 has not been executed.

AUTHOR CONTRIBUTIONS

TAS and AA conceptualized the study and developed the research methodology. RMA supervised the study and provided guidance on the research design and interpretation of the findings. MTN and AA contributed to the investigation, data analysis, and validation. TAS and AA conducted the implementation, experiments, and visualization. TAS prepared the initial manuscript. All authors contributed to the revision and editing of the manuscript, critically reviewed the final version, and approved the submitted version.

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CONFLICT OF INTEREST

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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